



Mobile Phone Data for Policy Program

UN 2.0



Featured Speakers



Esperanza Magpantay

Senior Statistician, ITU



Sveta Milusheva

Senior Economist,
World Bank



Cathy Riley

Director, Flowminder



Bruna Calazans

Director, Nommon



Trevor Monroe

Senior Program
Manager, World Bank

gdfmpd@worldbank.org



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GDF GLOBAL DATA FACILITY

MOBILE PHONE DATA FOR POLICY



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GOBIERNO
DE ESPAÑA

MINISTERIO
DE ECONOMÍA, COMERCIO
Y EMPRESA





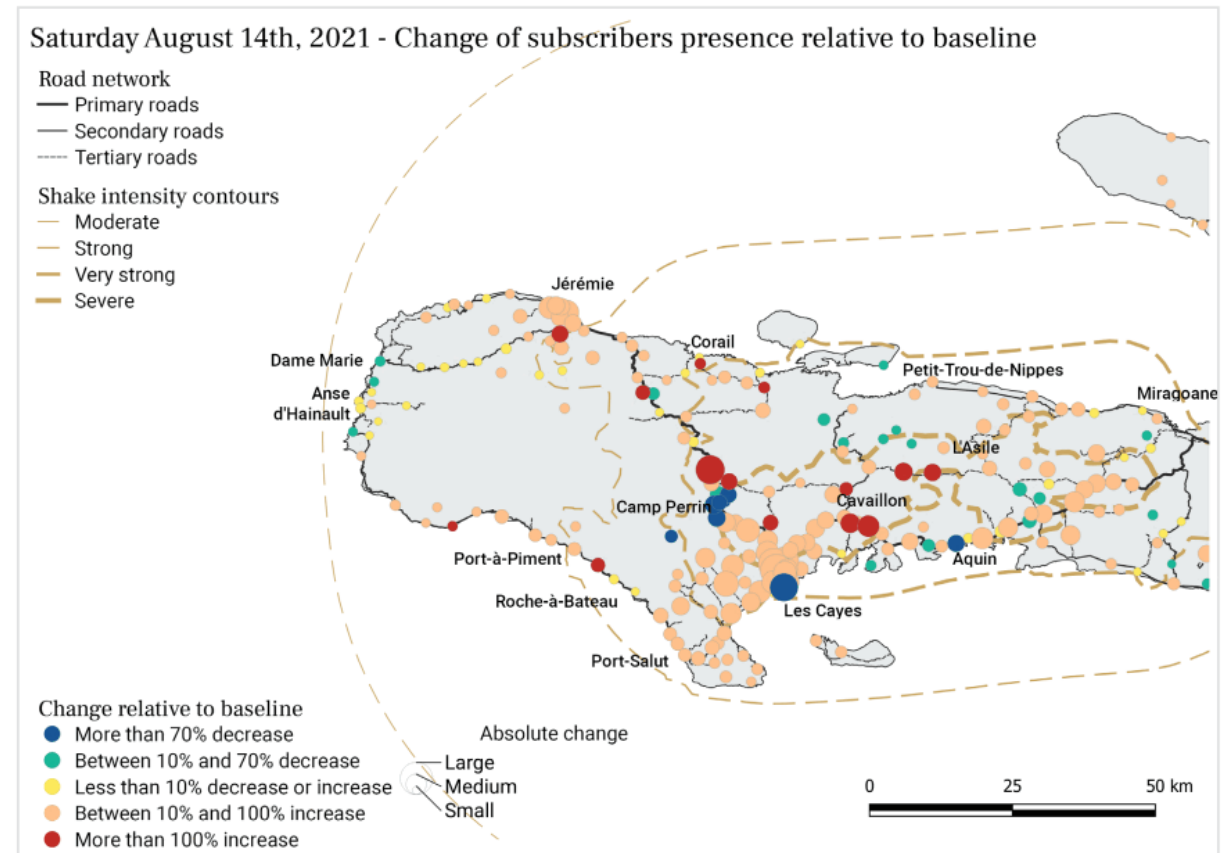
Mobile phone data can be a
powerful tool for better policy
to accelerate the SDGs



SDG 13: Climate action

More effective disaster response by knowing where people are

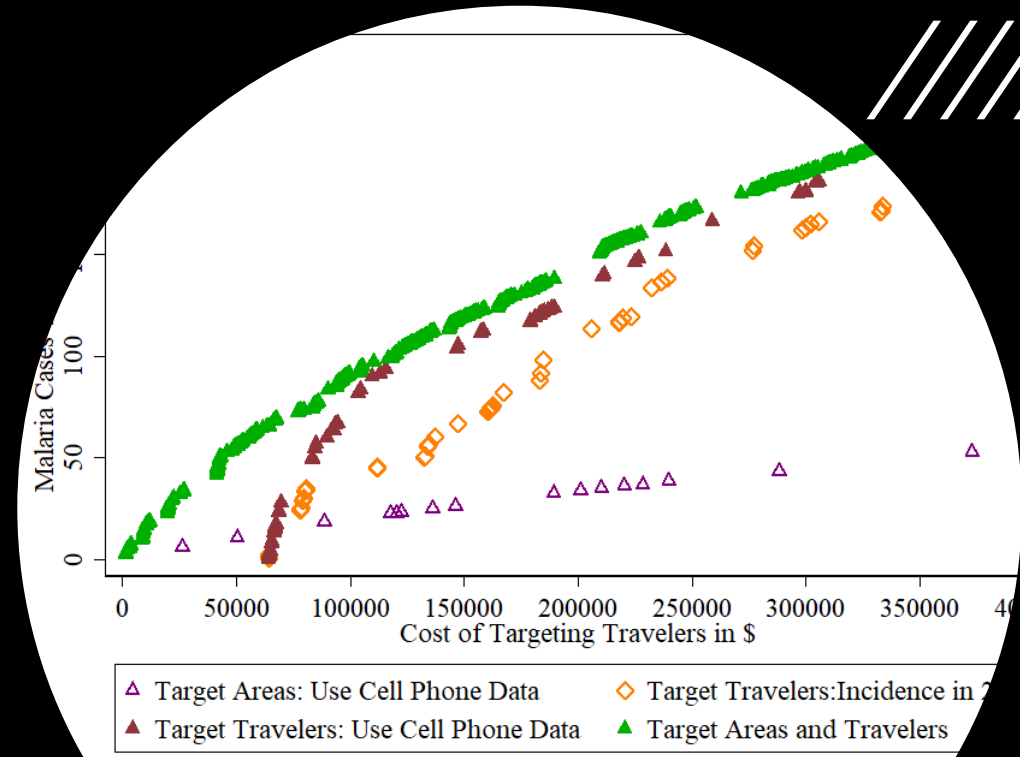
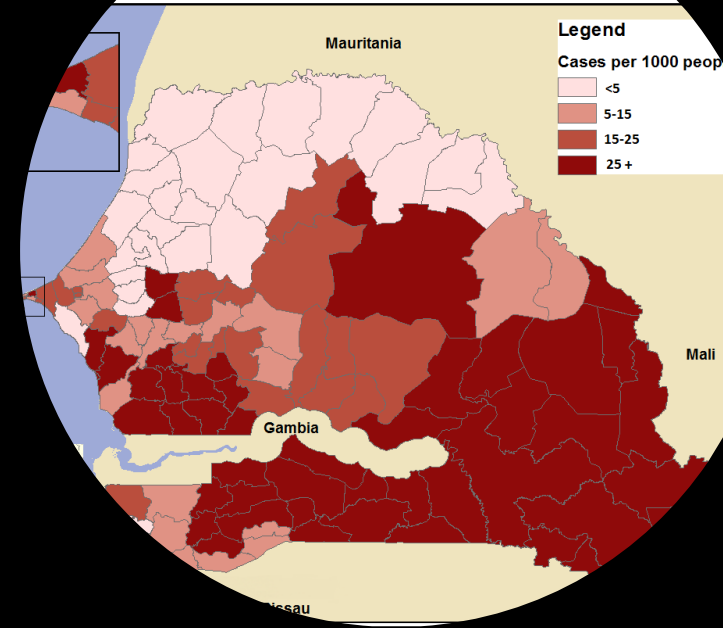
Population Change in Wake of 2021 Haiti Earthquake



Source: Flowminder Foundation. "Population movements estimated with mobile operator data from Digicel Haiti." 20 August 2021.

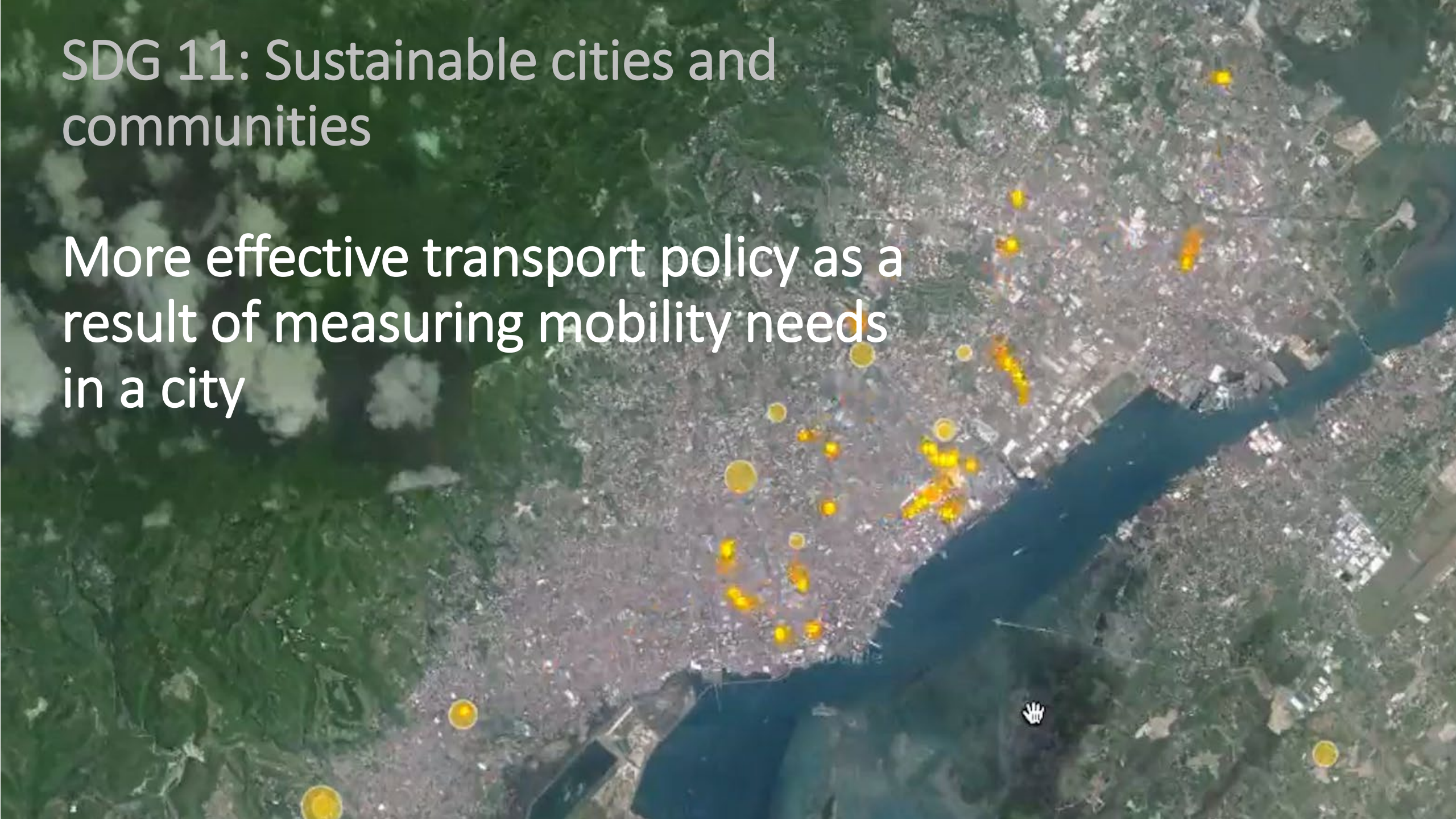
SDG 3: Good health and well-being

More effective disease prevention as a result of better targeting of resources



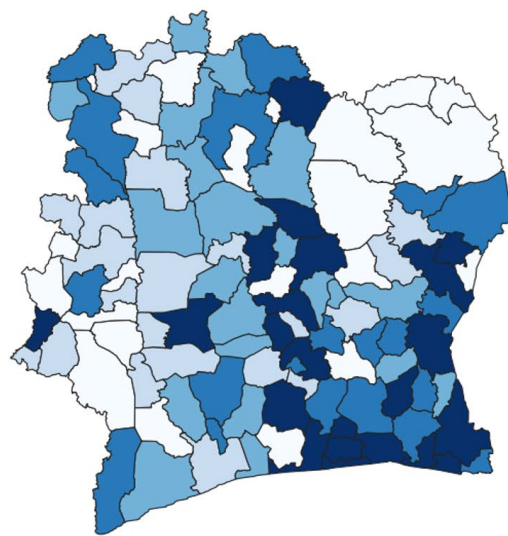
SDG 11: Sustainable cities and communities

More effective transport policy as a result of measuring mobility needs in a city



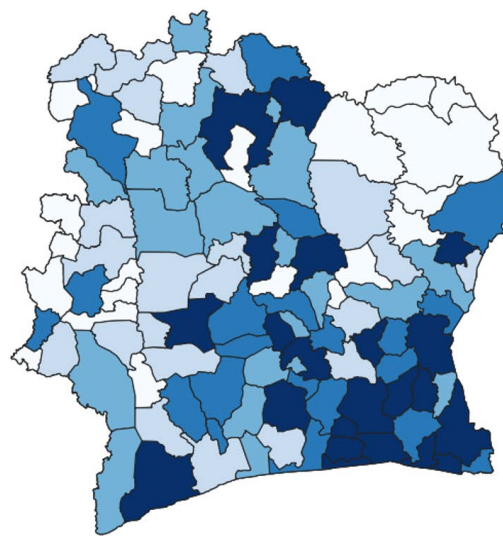


SDG 1: No poverty Improved mapping of poverty



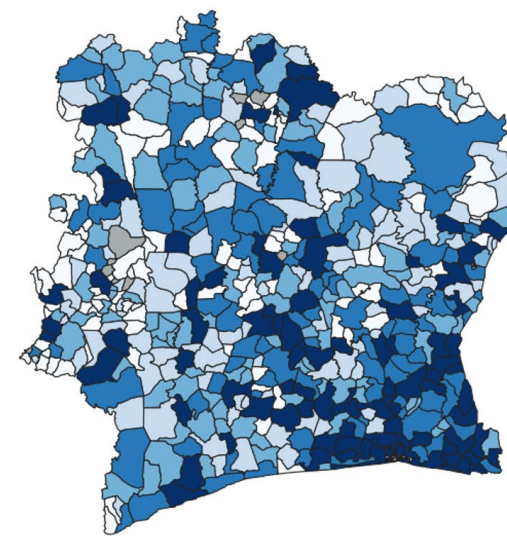
(a)

Survey Data



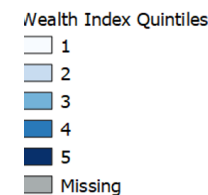
(b)

Prediction Using
Mobile Phone Data



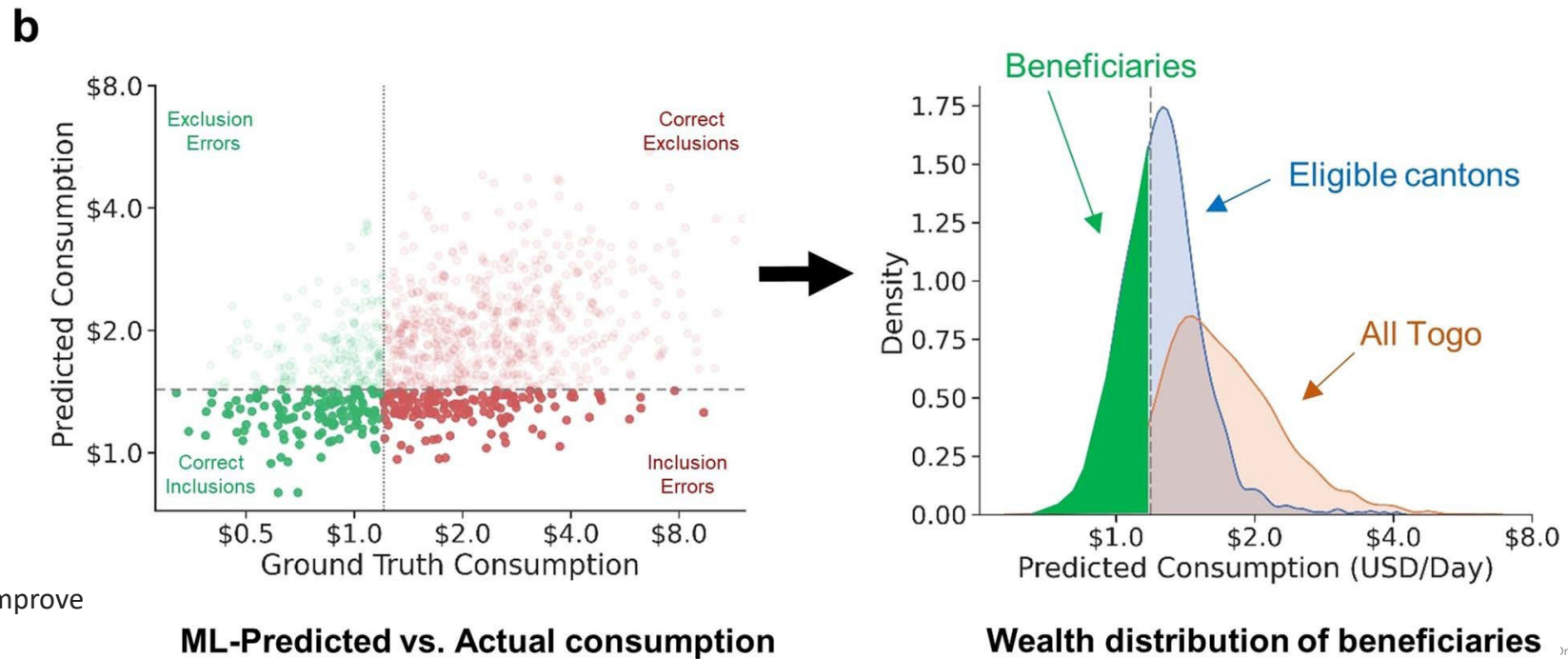
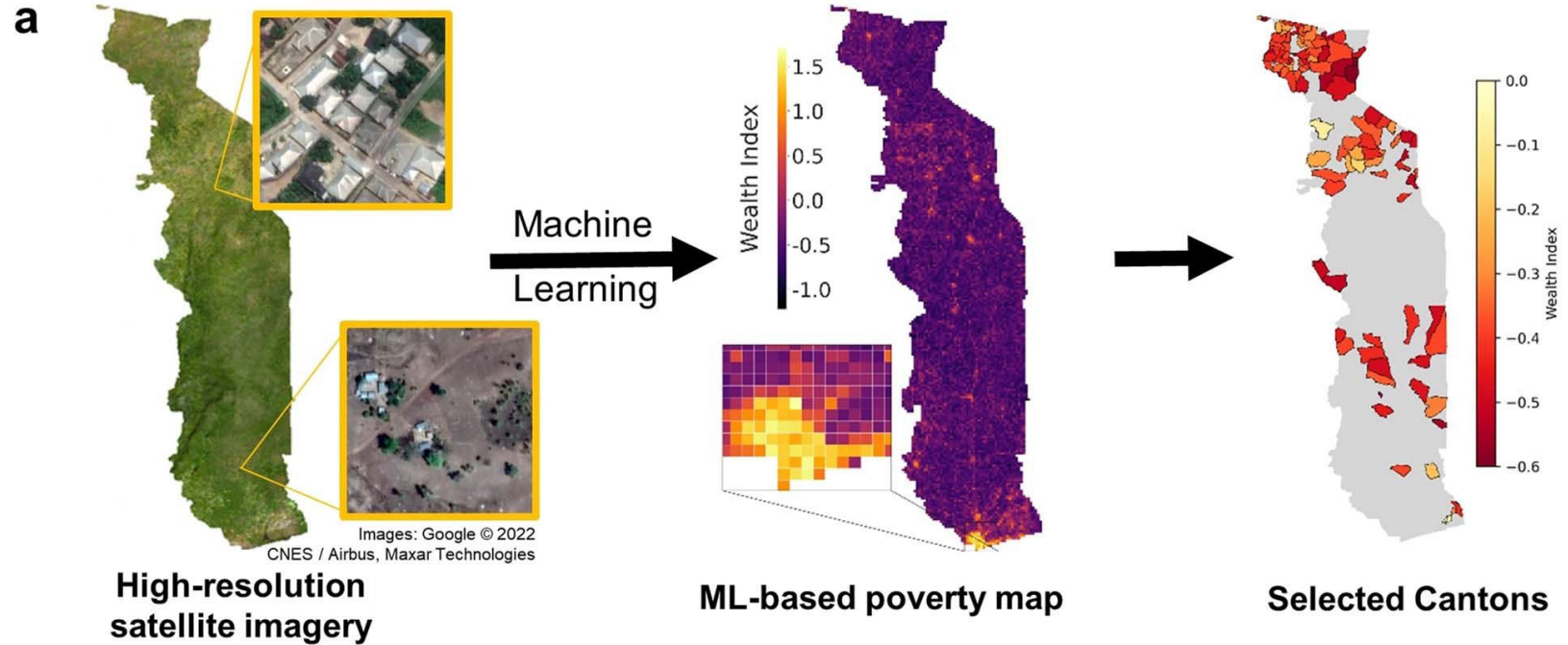
(c)

Prediction Using
Mobile Phone Data



SDG 10: Reduced inequalities

More effective social protection programs due to fast targeting during crises



Challenges





Opportunity

UN 2.0 can leverage mobile phone data to improve policies and targeting for greater SDG impact



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Mobility data for disaster preparedness & response

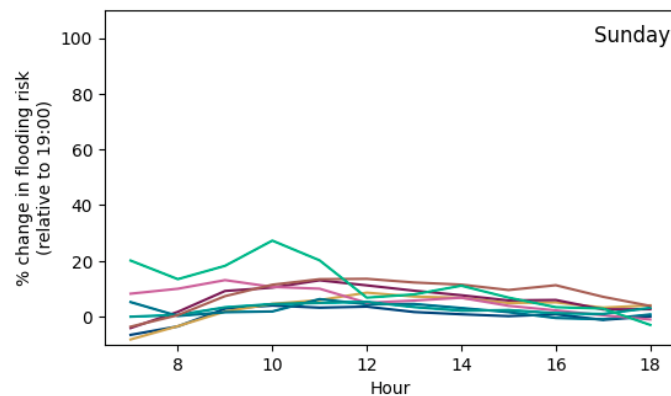
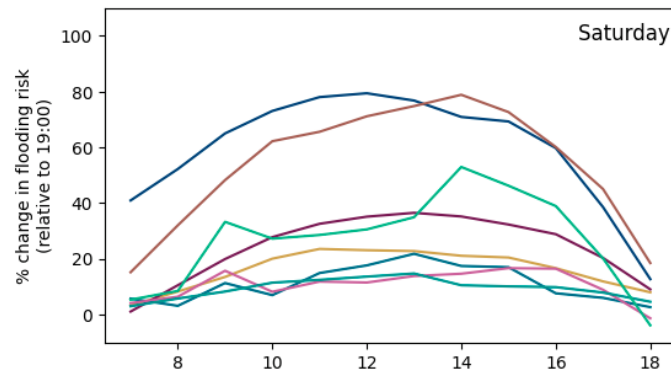
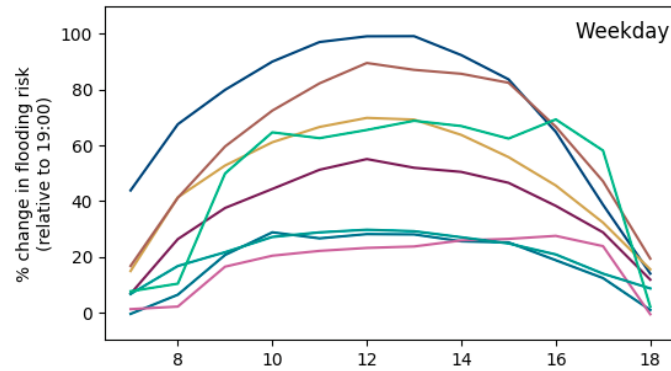
Disaster management stages and data

1 Preparedness (Baseline)	How many people live within a disaster affected area? How many people are in specific areas at specific times? How many people are connected and reachable by phone?
Hazard Forecast	How many people may be affected by a specific hazard?
Hazard Warning	Are people moving or relocating as the results of warnings?
Active Crisis	How many people are displaced, and from where to where?
Protracted Crisis	How is the crisis (several events) impacting population distribution patterns?
Short-Term Recovery	Are people returning to their homes or staying at their displaced locations, and how long is recovery taking?
Long-Term Recovery	Are resettlement areas growing or depleting over time (in terms of months and years)?

 Decision-making occurs throughout the Disaster Management Cycle

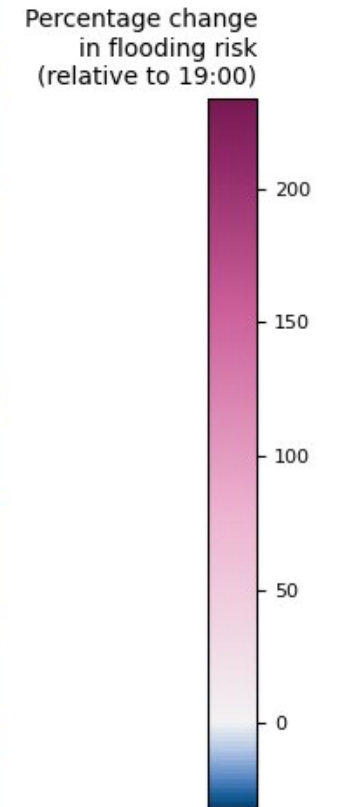
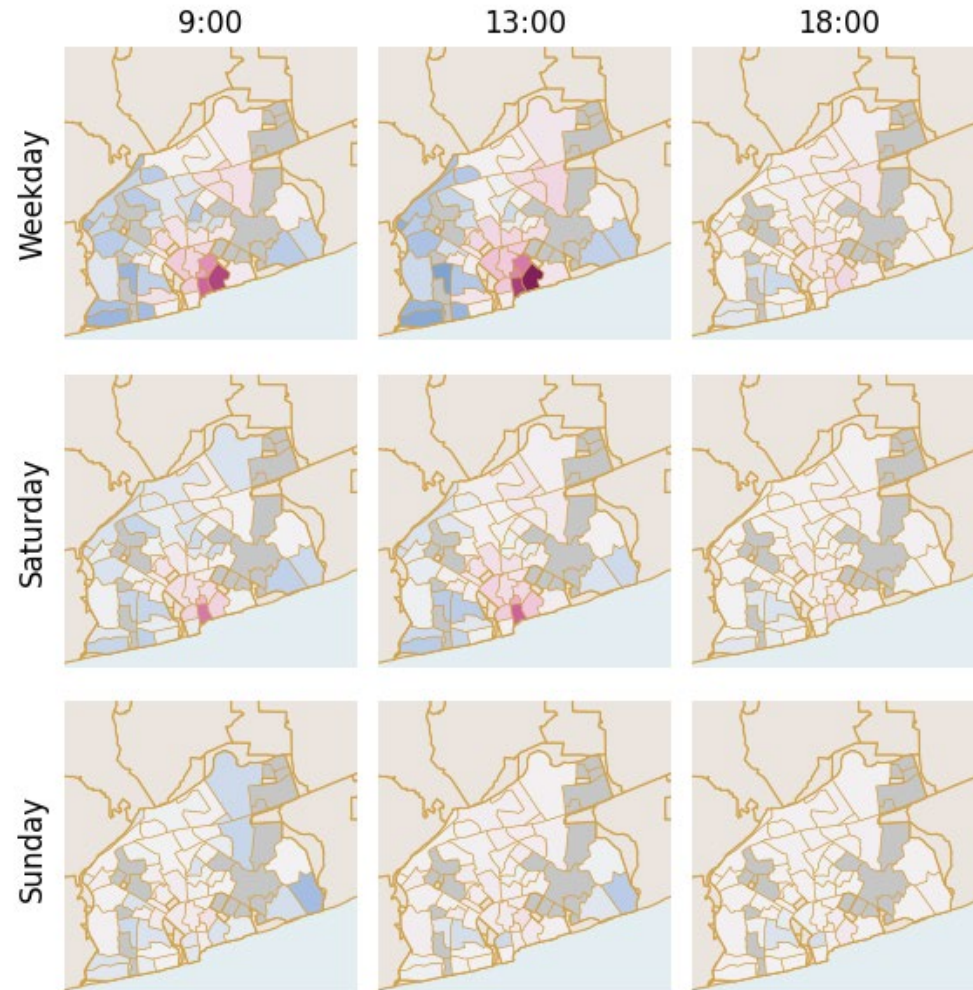
Analysis of mobile phone data can provide insights to support those decisions at different stage

Dynamic risk forecasting | Flood exposure risk - Accra, Ghana



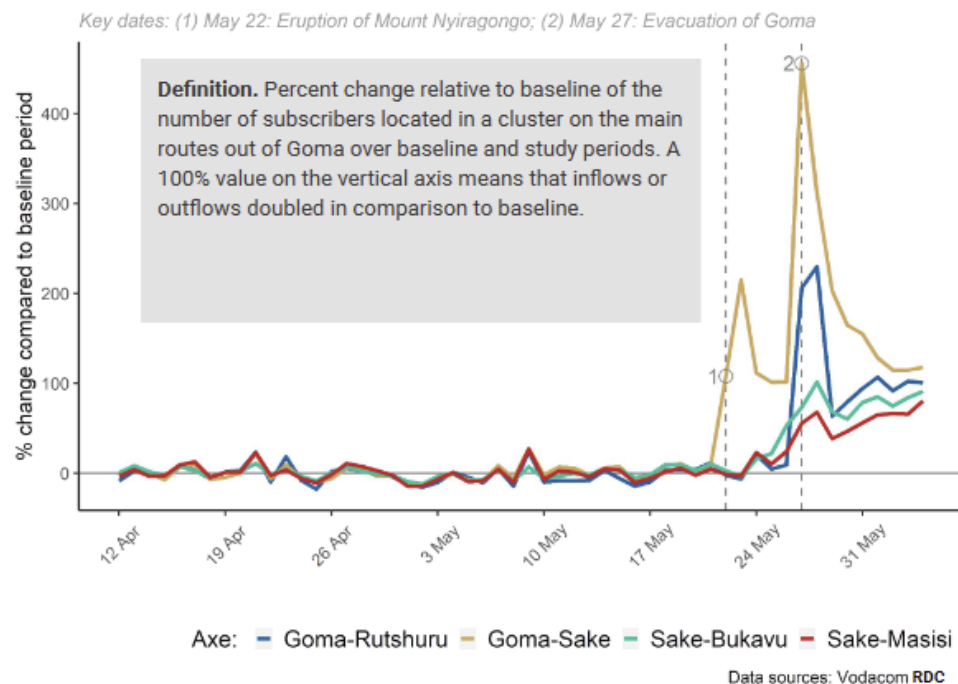
District code

- fid025
- fid031
- fid127
- fid155
- fid179
- fid215
- fid241
- fid247



Hazard warnings | Volcanic eruption & evacuations - DRC

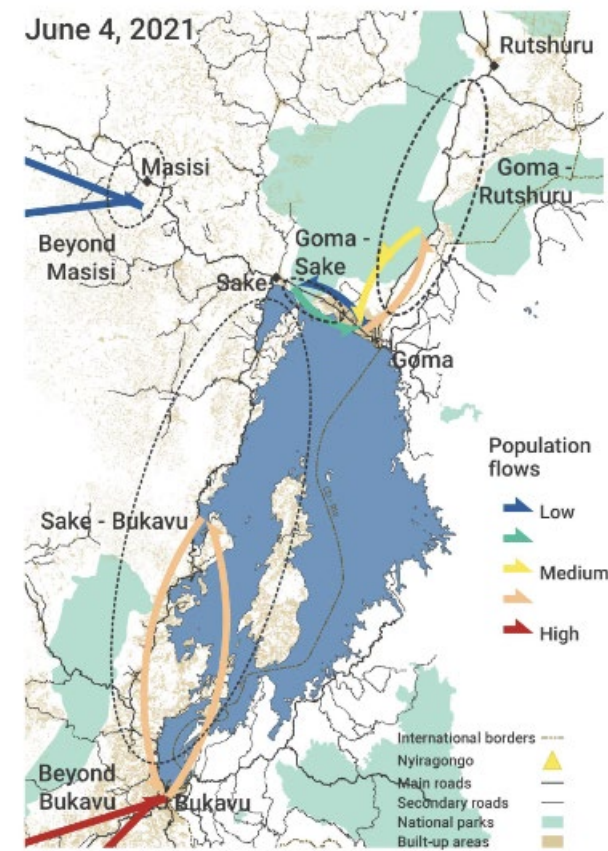
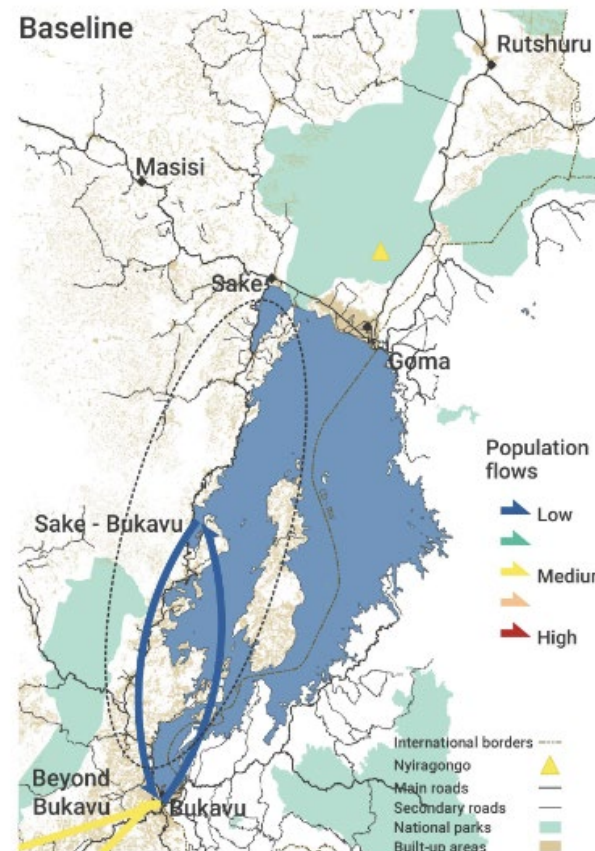
Subscribers presence on the main routes out of Goma



Flows of subscribers

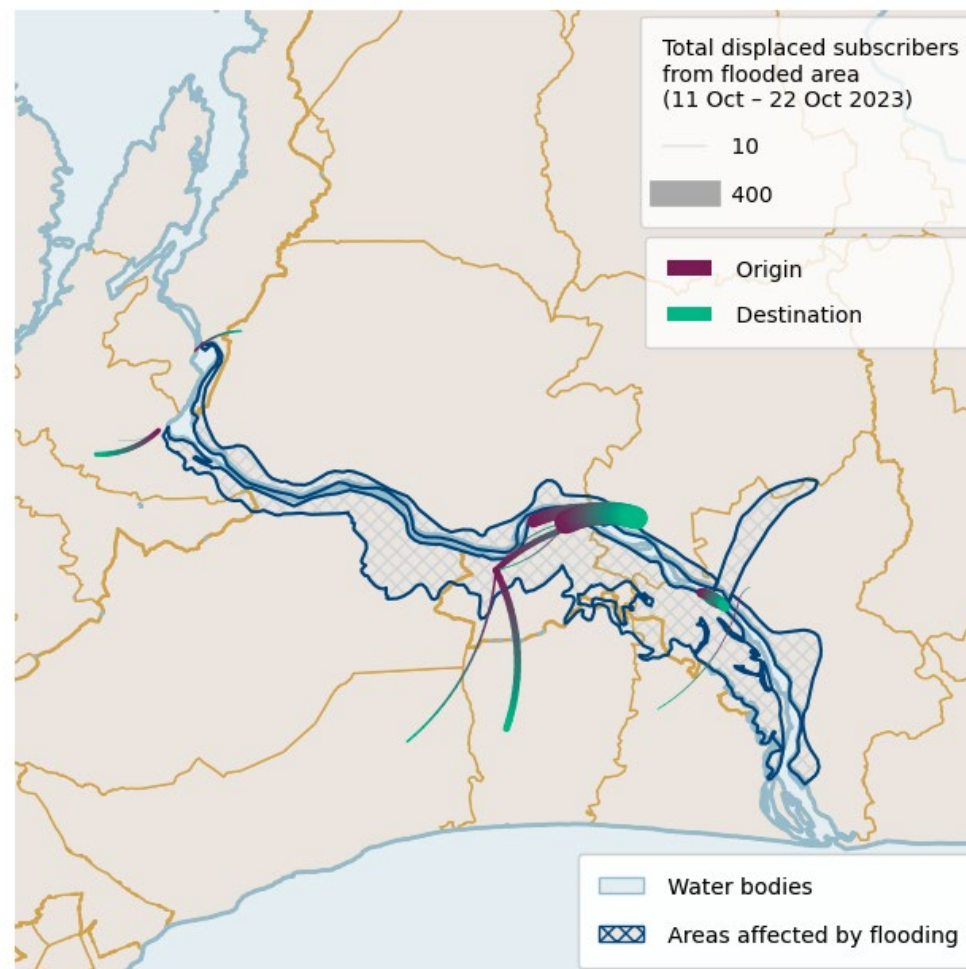
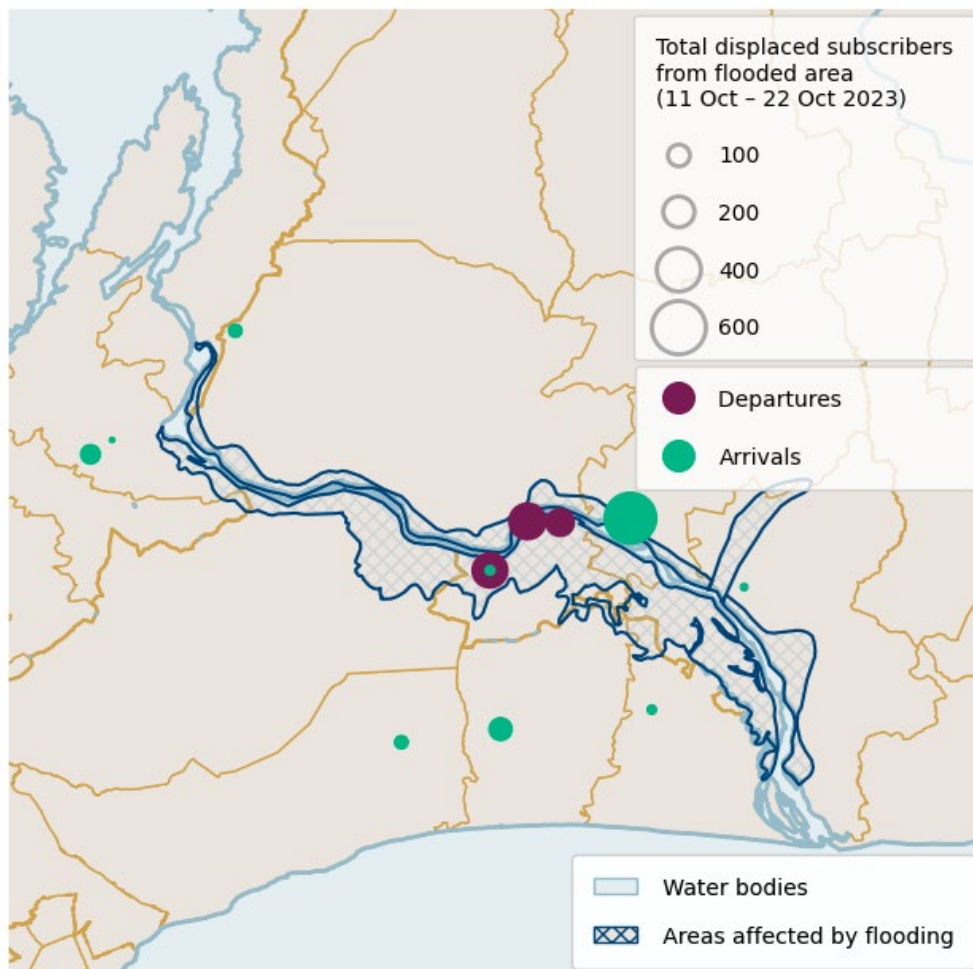
Baseline (22 May)

04 June



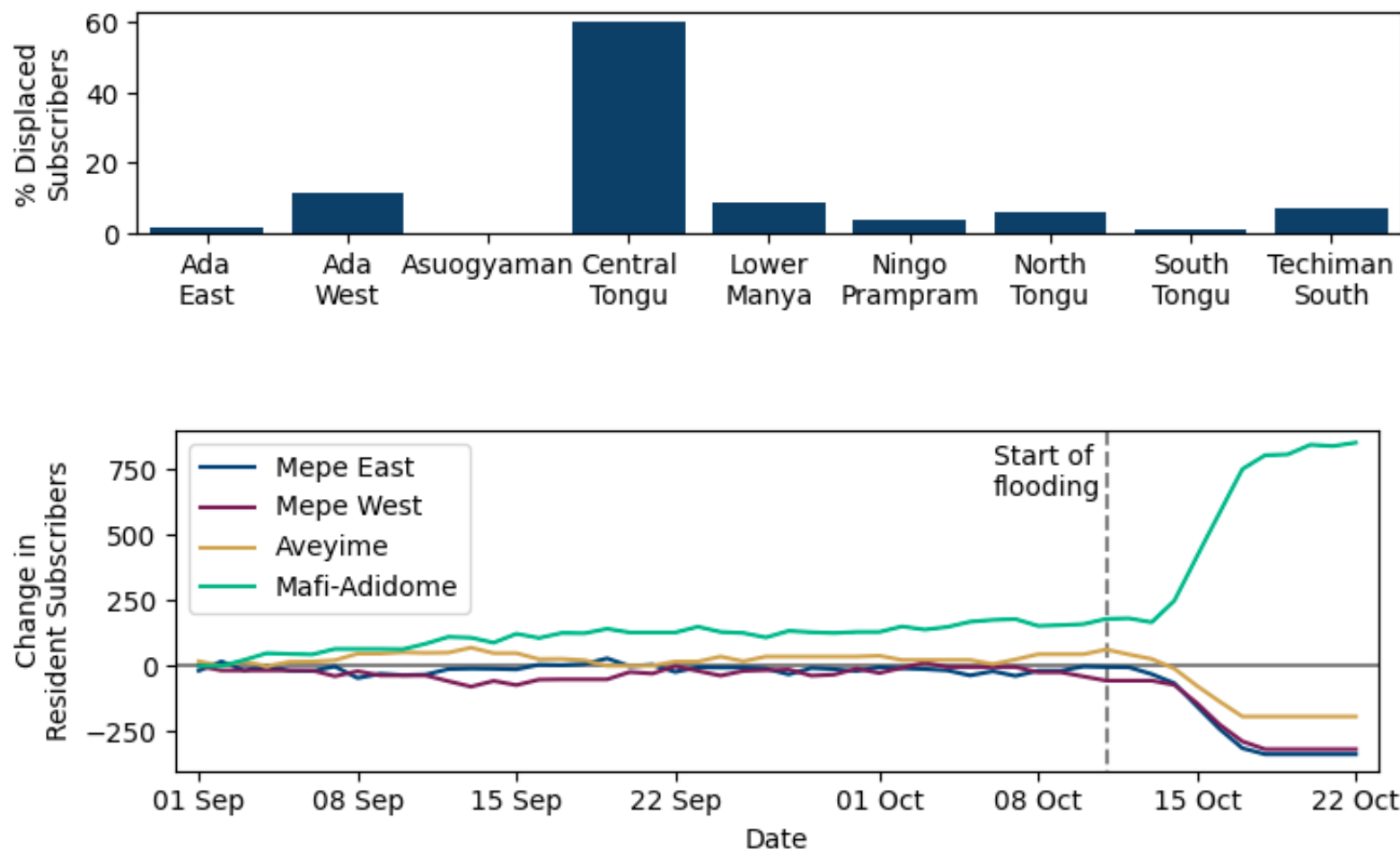
Estimating population displacement following the Eruption of Mount Nyiragongo, DRC (22 May 2021)

Active crisis | Flooding displacements - Lower Volta, Ghana



Active crisis | Flooding displacements - Lower Volta, Ghana

- Estimate the number of subscribers displaced to and from different areas
- Estimate the share of displaced subscribers in each destination location
- Provide insights on when displaced subscribers return to their previous residence locations



“

The data told us where some of the **displaced persons** had moved to.

— Kelly Kekely Degboe, Head of Emergency Operations Centre (EOC) & Information Technology (IT) at **NADMO**.

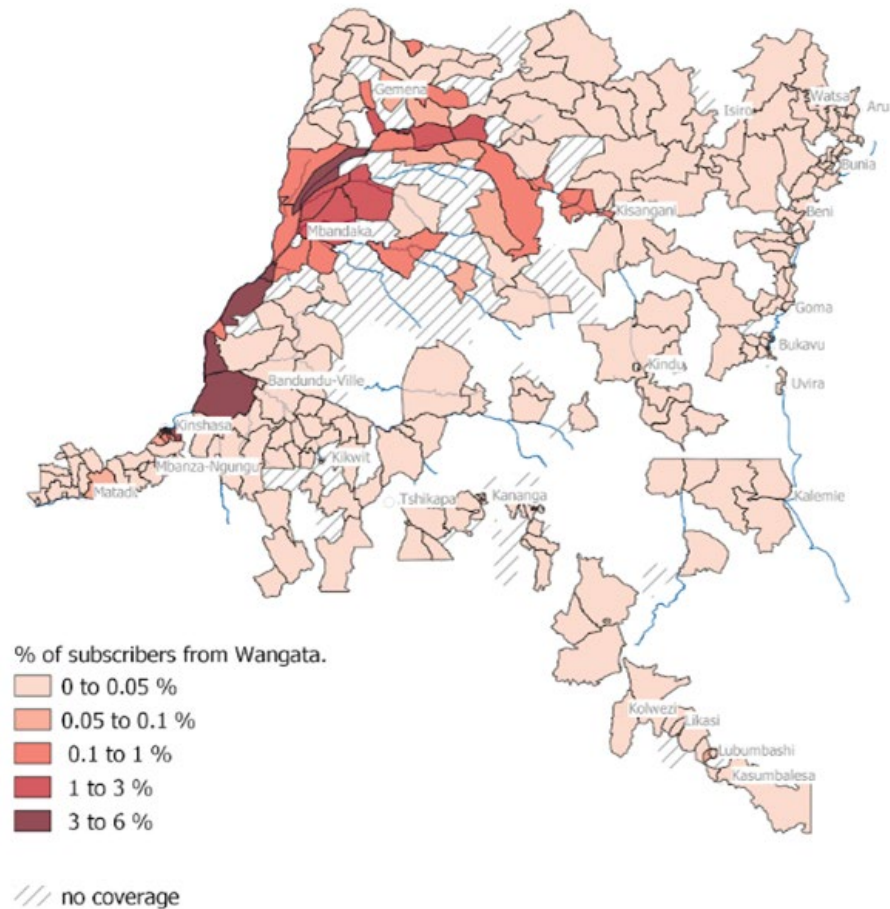
Watch the full story



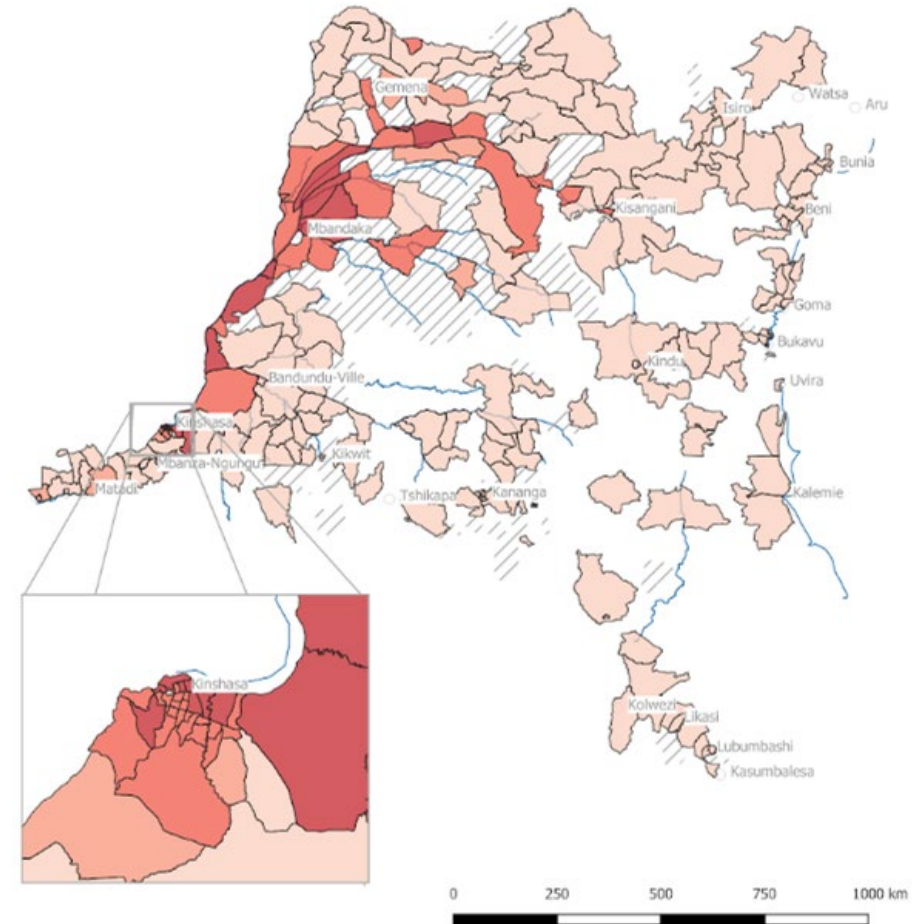
FLOWMINDER.ORG

Active crisis | Disease spread modelling - DRC

Zones visited by subscribers **before** appearing in the Ebola area



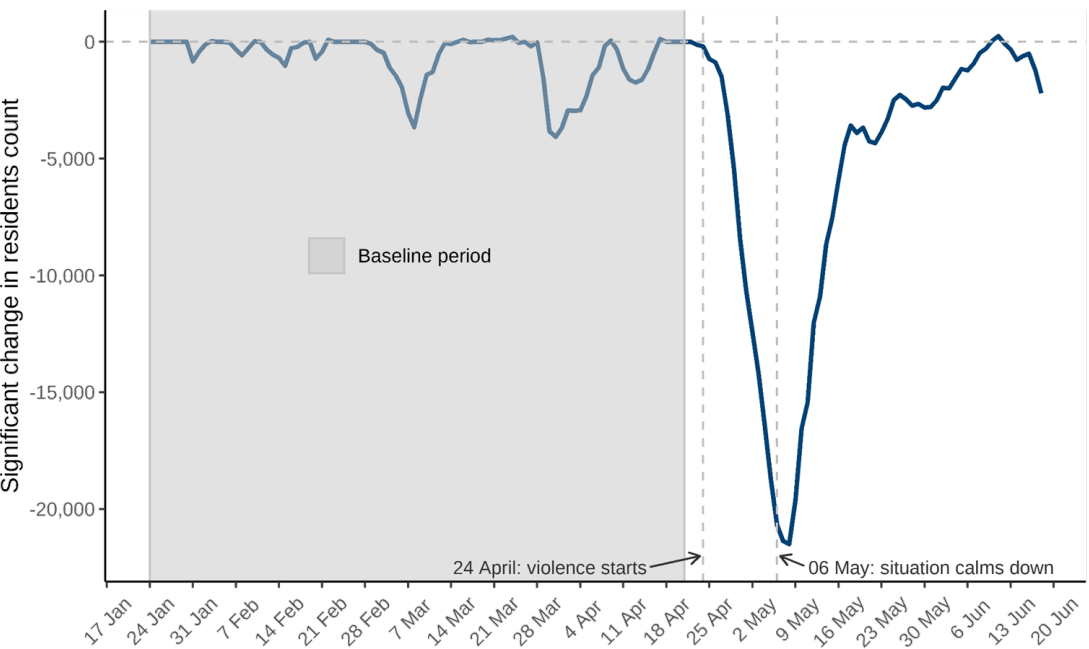
Zones visited by subscribers **after** appearing in the Ebola area



Protracted crisis | Haiti - Gang violence displacements

Significant change in the count of Digicel residents within the affected area

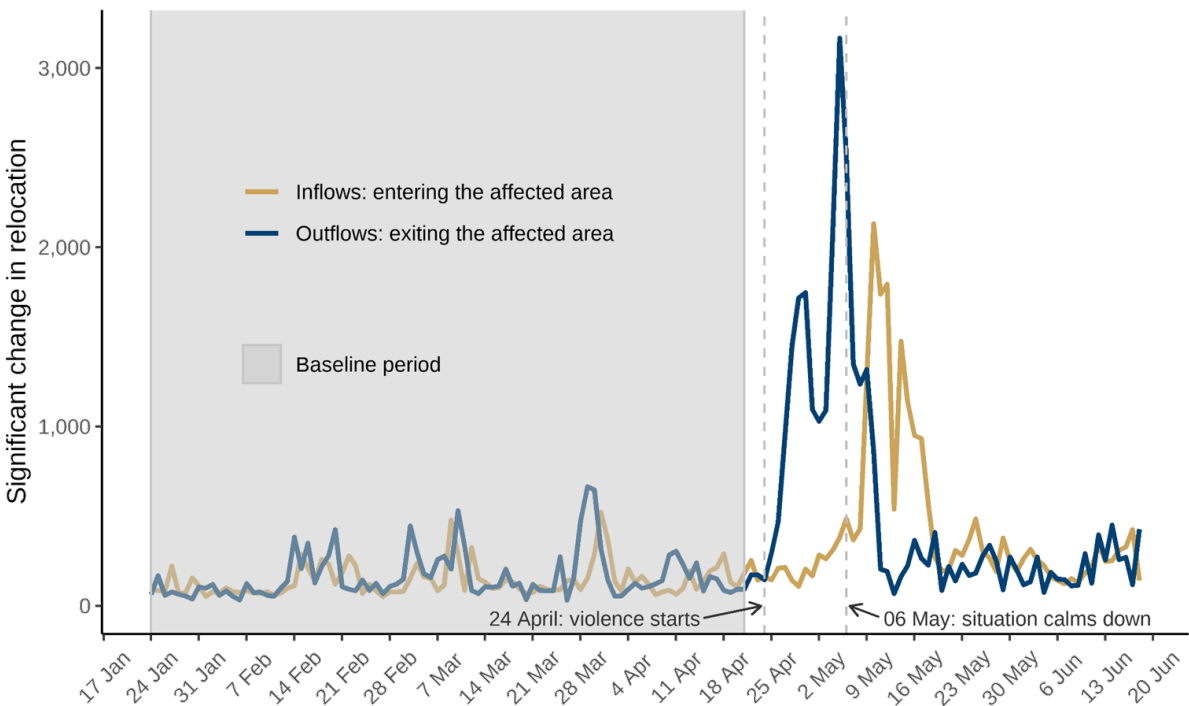
Digicel residents: active Digicel SIM cards with modal home location within the affected area during the baseline period.



Data source: Digicel Haiti

Significant change in the count of Digicel residents relocating from/to the affected area

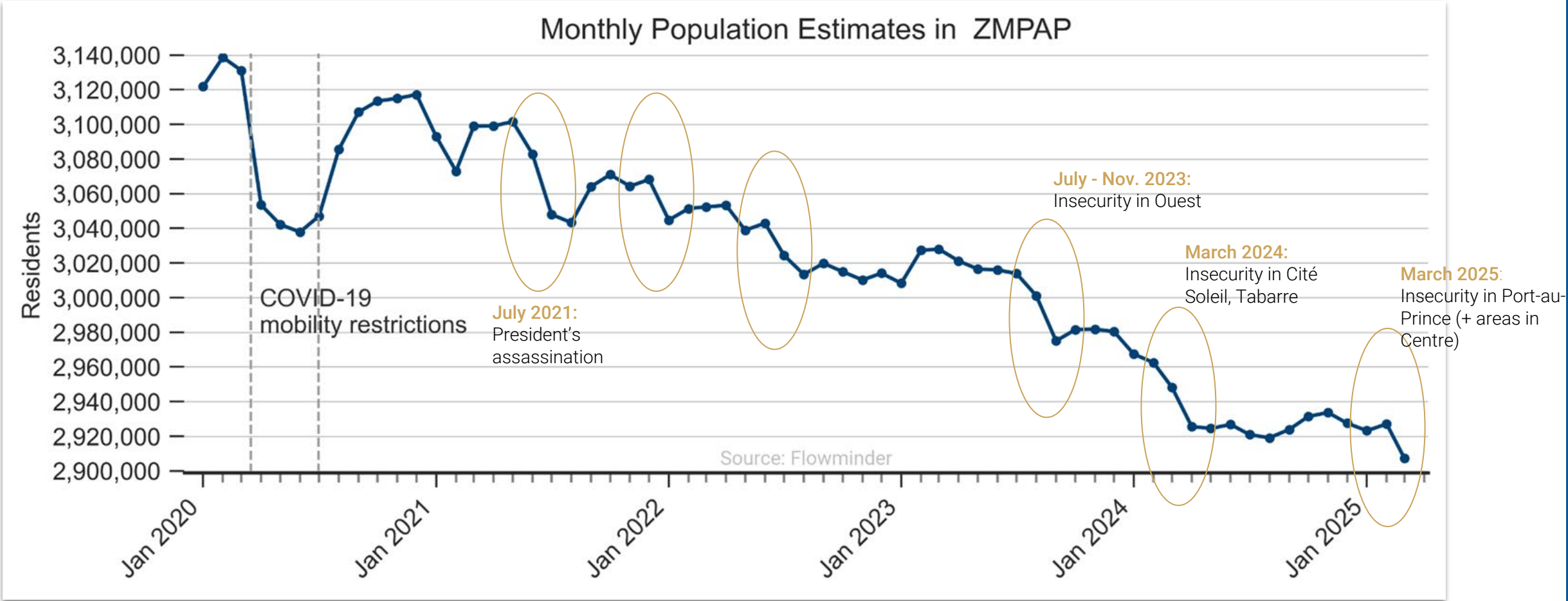
Relocations: active Digicel SIM cards with a change of home location



Data source: Digicel Haiti

Gang violence in Plaine du Cul-de-Sac, Haiti (April-May 2022)

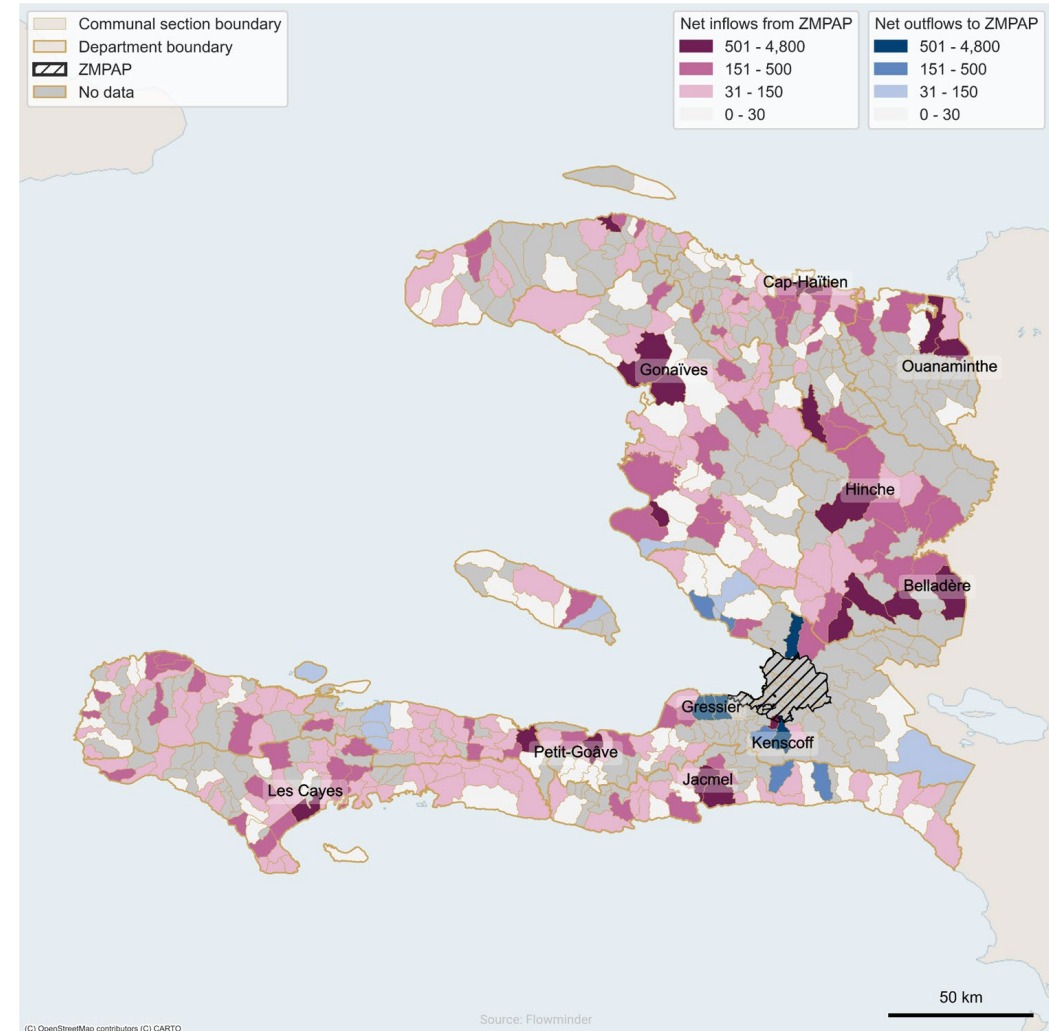
Protracted crisis | Haiti - Multiple displacements over time



Protracted crisis | Haiti - Multiple displacements over time

The national scale: Where are people going if leaving the capital, Port-au-Prince?

- Estimated population change from mobility with PaP Metro Area per communal section, April 2024 to March 2025
- In the past 12 months: net inflow from PaP Metro Area to most communal sections in Haiti



“

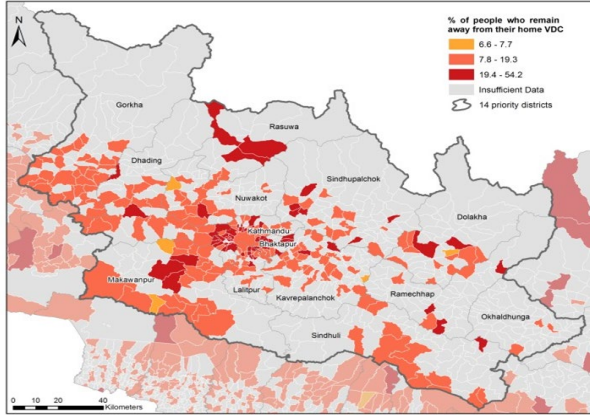
The data made it clear that Port-au-Prince was emptying...it confirmed our hypothesis about the **population decline** and, crucially, provided the numbers to support it.

— Jean-Martin Bauer, WFP Country Director for Haiti

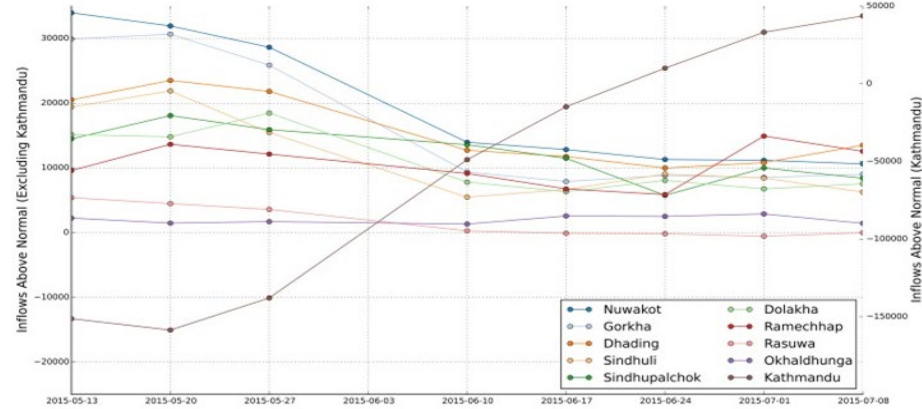
Read more online



Recovery | Rates of return to affected areas



Percentage of people not returned
on 19 August 2015



Anomalous inflows per districts

Nepal Earthquake, April 2015

Total displaced
subscribers
(‘remaining’ displaced)*

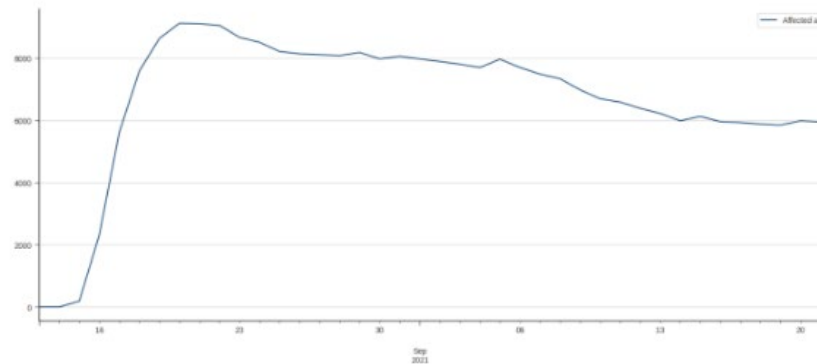
6,000

% of displaced
subscribers returned to
Les Cayes

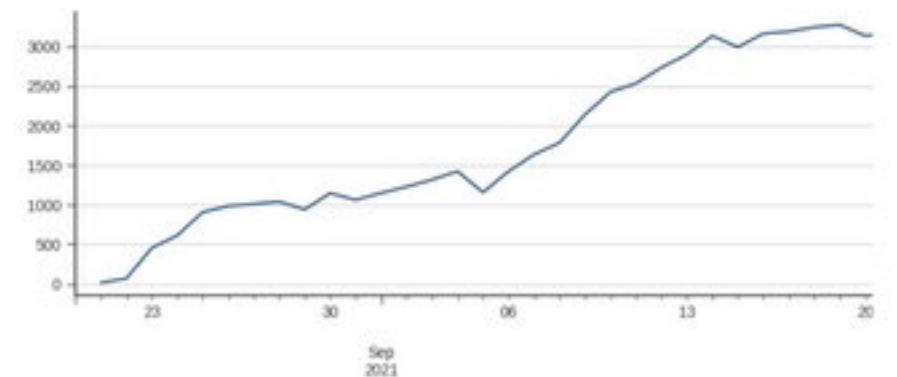
34%

Haiti
earthquake
2021

Evolution of the
total number of
displaced
subscribers



Evolution of the number of returnees to Les Cayes from known host locations



FLOWMINDER.ORG

Cathy Riley

info@flowminder.org



www.flowminder.org



info@flowminder.org



[@Flowminder](https://twitter.com/Flowminder)

NOMMON

MPD for Transport
< LAC >

Bruna Calazans

WHO WE ARE AND WHAT WE DO

- Founded in 2012, in Spain
- Technical team mostly in Spain, with additional support in the UK, Brazil, Colombia and México
- Over 300 projects completed

Information, models and decision support tools for the planning and management of transport and mobility systems

Descriptive analytics



Mobility Insights

Best-in-class origin-destination matrices from mobile network data



Population Insights

Leveraging mobile location data to understand visitor footfall and customer behaviour

Predictive and prescriptive analytics



WiseTransit

Passenger-centric optimisation of public transport services



WiseRide

Planning and managing shared mobility services through AI-powered demand modelling



WisePax

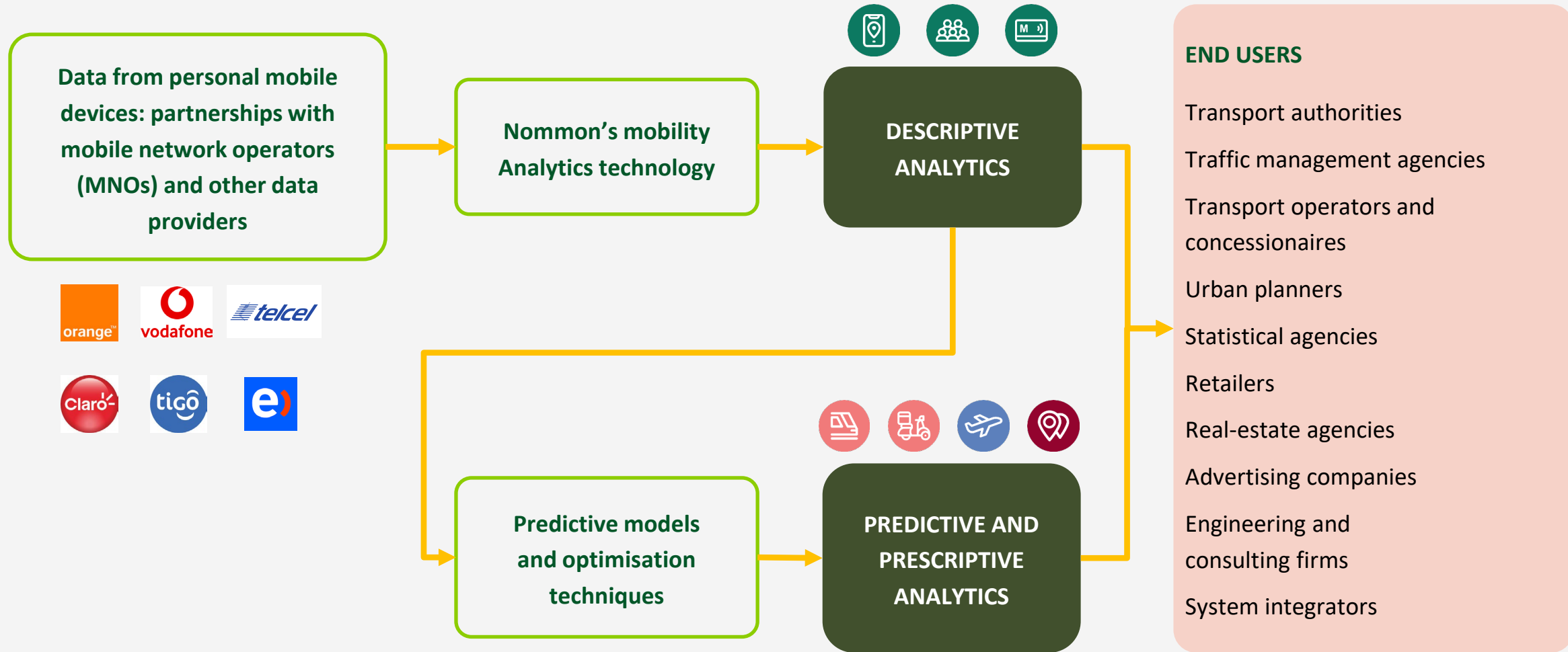
Passenger intelligence for the smart airport of the future



InSite

Transforming location intelligence through high-resolution mobility data and predictive analytics

HOW WE DO IT



GDF-MPD PROJECT

- Training on MPD for policy and statistics
- World Bank + Flowminder + Positium + [Nommon](#) + University of Tokyo
- Jan 2025 – Jun 2026
- Online and onsite trainings
 - Data Governance, Protection, Privacy, and security
 - Considerations for Multi-MNO data pipelines
 - Case studies

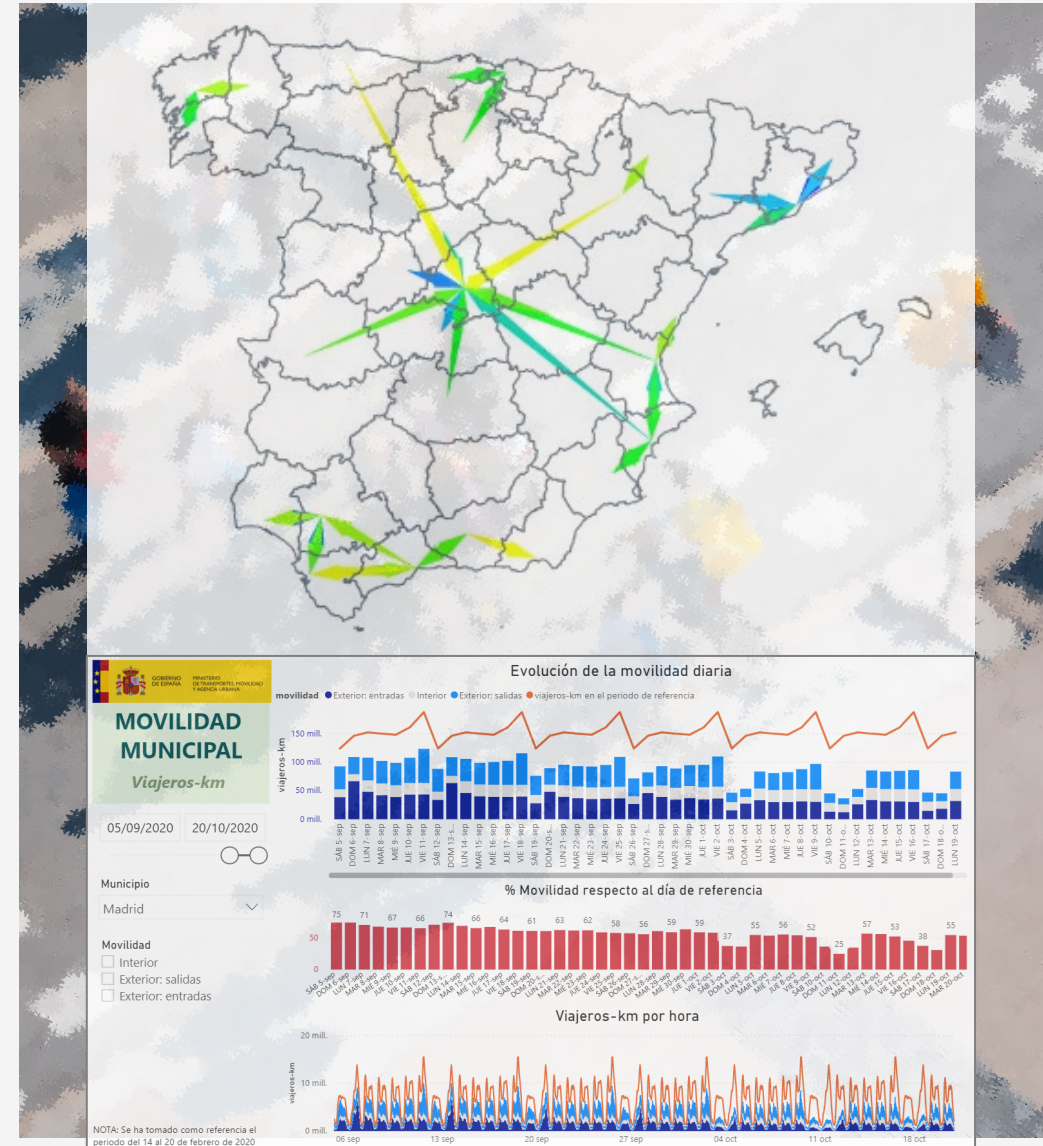




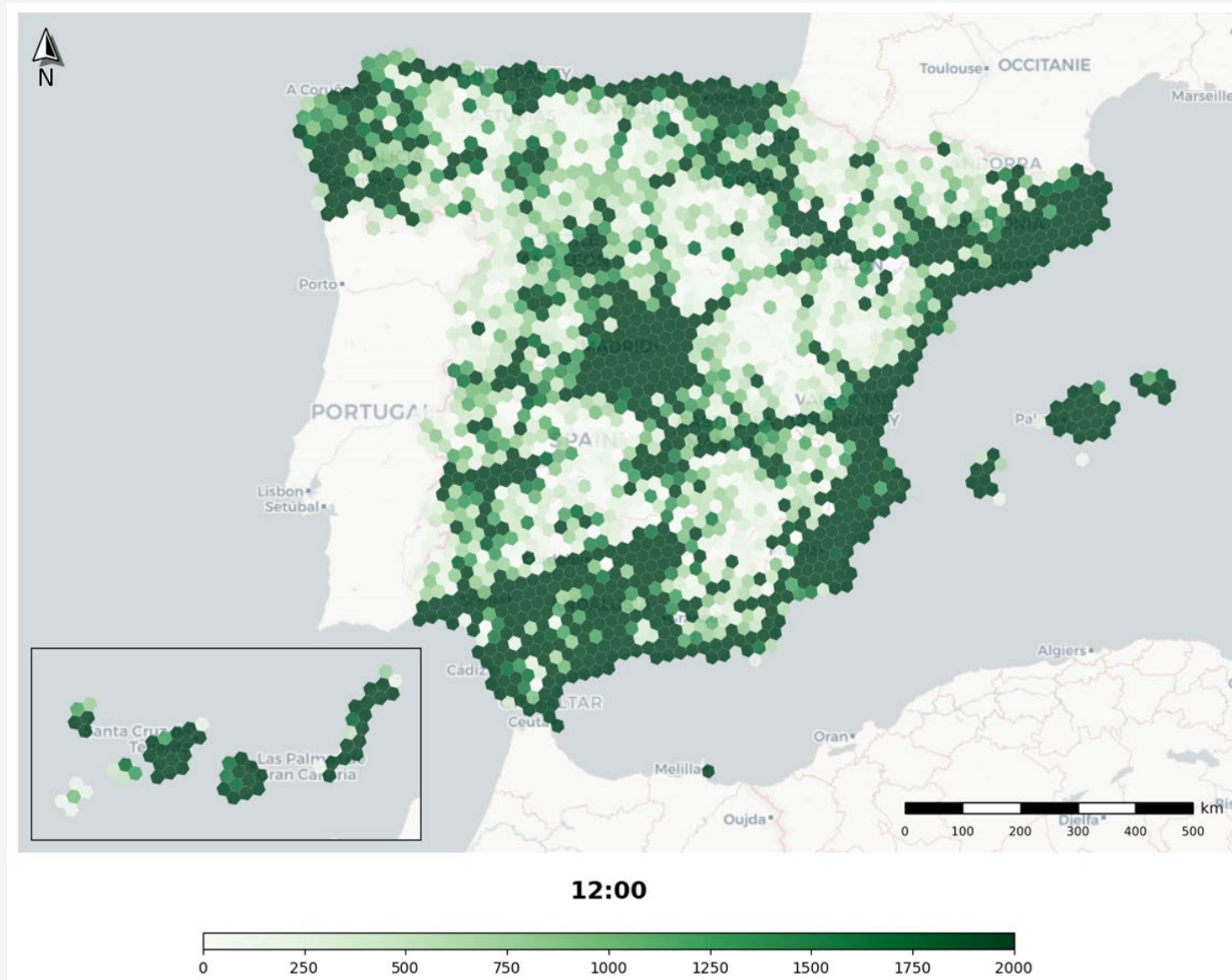
Europe experience

CONTINUOUS MOBILITY ANALYSIS

GOALS 	Daily analysis of mobility flows, monthly indicators per mode, and semi-annual information on road route choice
CLIENT 	Project promoted by the Spanish Ministry of Transport, extension of a previous contract for pandemic analysis studies
CHALLENGES 	– National-level modeling of residents and foreigners. – Continuous monitoring with a 3-4 day response time. – Consolidation of historical databases, enabling the study of the evolution of mobility patterns



CONTINUOUS MOBILITY ANALYSIS



- ✓ Monitoring events
- ✓ Power outage in Spain on April 20, 2025
- ✓ The power loss was gradual, beginning at 12:00
- ✓ The disconnection was not uniform across all of Spain; large regions such as Madrid, Bilbao, and Barcelona maintained partial connectivity

REFERENCE FRAMEWORK FOR STATISTICS

OBJECTIVE 	Modeling calculation parameters and statistical procedures for the use of MPD in the European Union
CLIENT 	Project promoted by the Statistical Office of the European Union, extension of a previous contract for pandemic analysis studies
CHALLENGES 	– Multi MNO Project – Open data: framework + code – Revisited concepts/methods



A photograph of an airport terminal interior. The scene is captured from a low angle, looking up at a series of large, dark rectangular panels that form the ceiling or upper walls. Below these, a bright green horizontal band spans the width of the image. In the background, through large glass windows, the silhouettes of several people are visible. One person in the center is standing next to a red suitcase. The floor is highly reflective, mirroring the silhouettes and the bright light from the windows. The overall atmosphere is one of a busy, modern transit hub.

LAC experience

LAC PRESENCE

- Heterogeneous regions, with countries in different levels of maturity

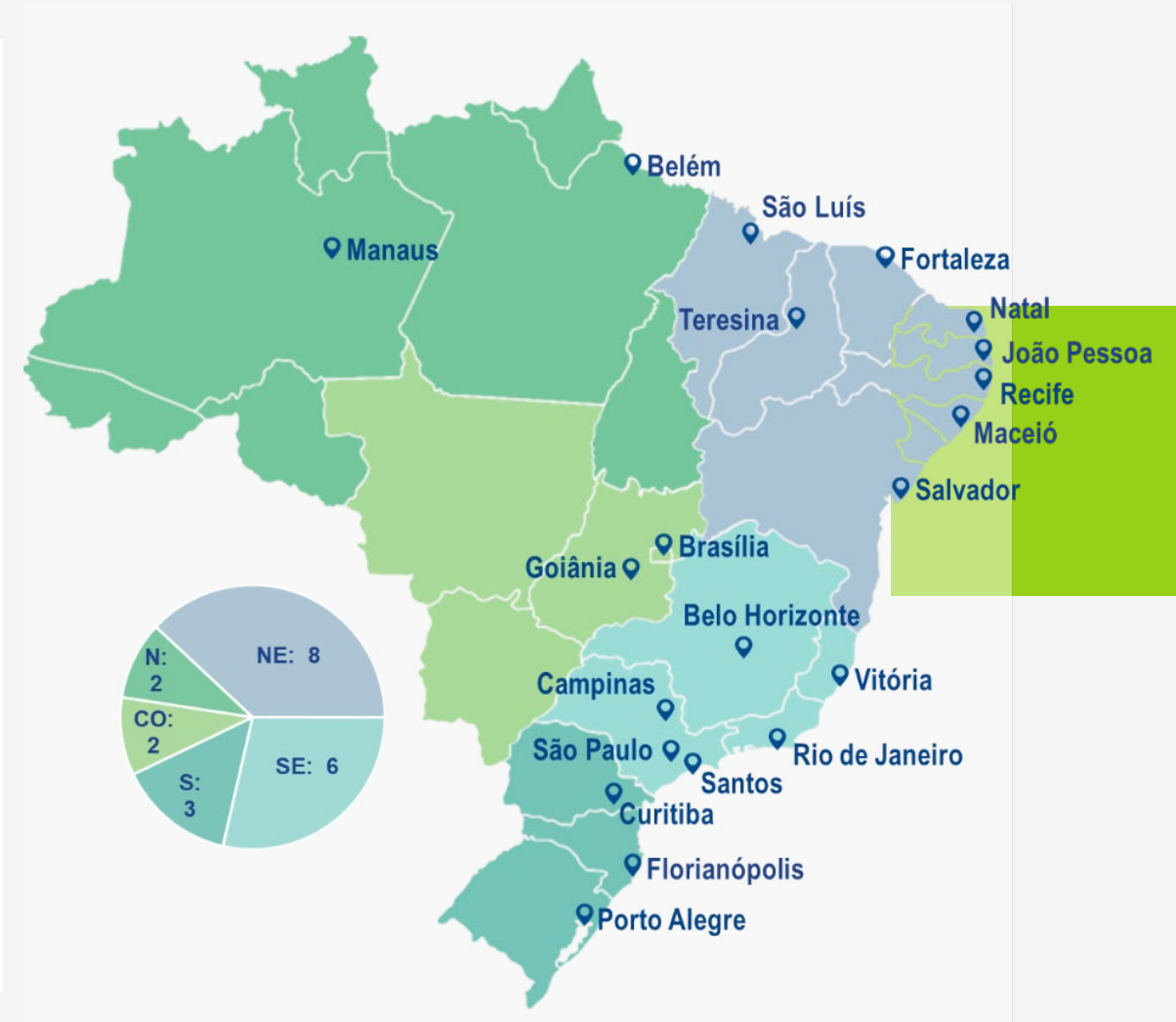
- Brazil
- Colombia
- México
- Chile
- Argentina/Paraguay/Uruguay

+MATURITY = +PROJECTS



21 Metropolitan Regions – BNDES Initiative (BR)

OBJECTIVE 	Generation of origin-destination matrices in 21 metropolitan regions of Brazil
CLIENT 	Brazilian consultancies studying the regions for a larger project for BNDES
CHALLENGES 	Urban study with national scope Short time frame for generating all matrices → calculation optimization Development of complementary indicators. - Immobility - Coverage análisis - TGR by coverage radius area



MOBILITY ANALYSIS WITH MODE SALVADOR (BR)

OBJECTIVES



Systematization of origin-destination matrices using MPD in Salvador (BA/Brazil). Studies of a typical week and a festive week (Carnival)

CLIENT

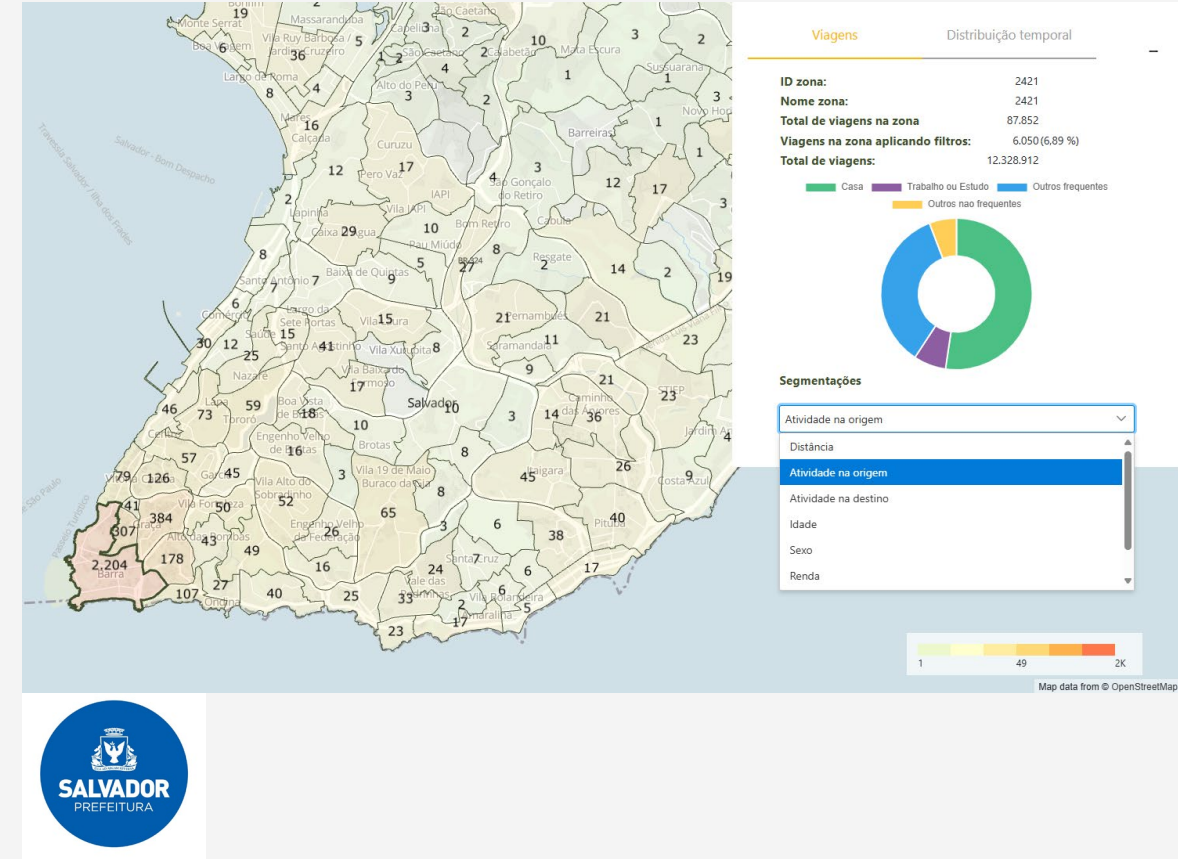


Municipal Secretariat of Urban Mobility

CHALLENGES

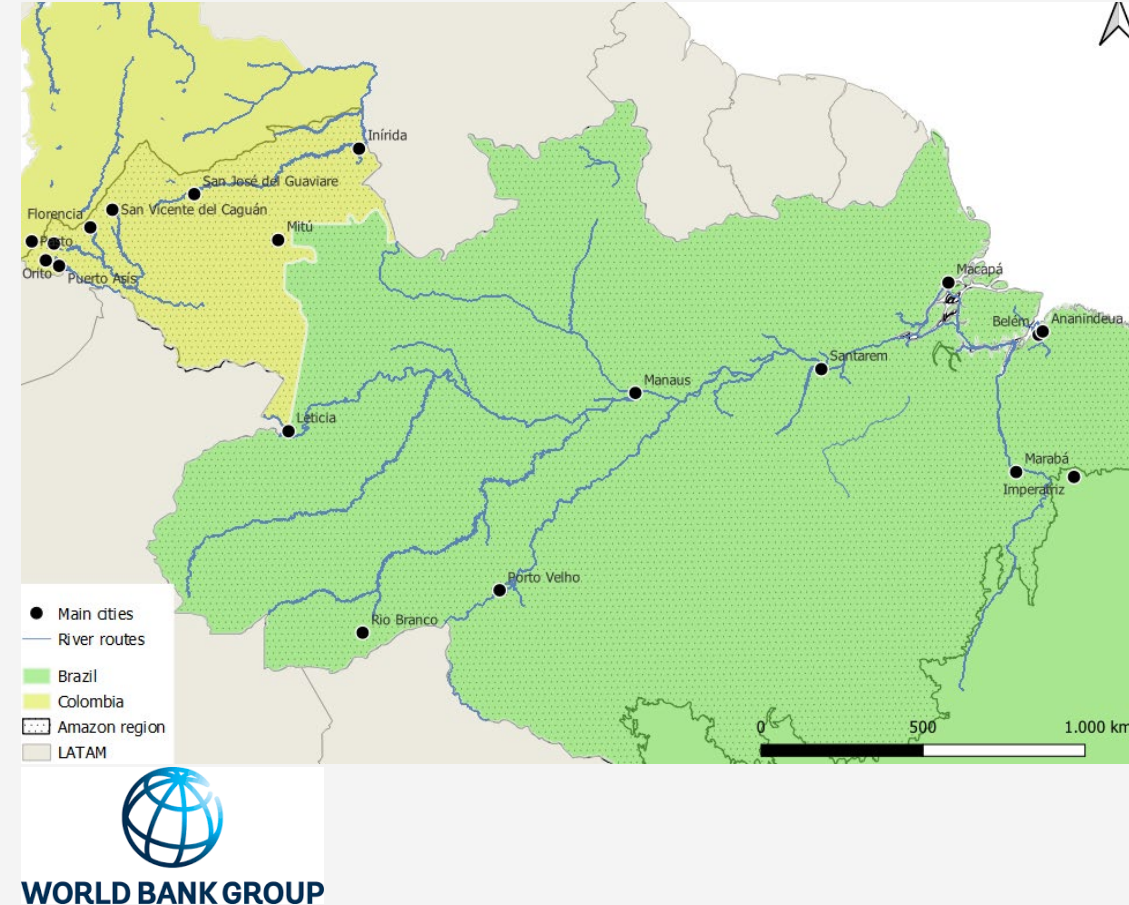


- Collection of electronic ticketing data for construction of the mode prediction model.
- Development of an analytical tool for data exploitation with a simple and intuitive interface



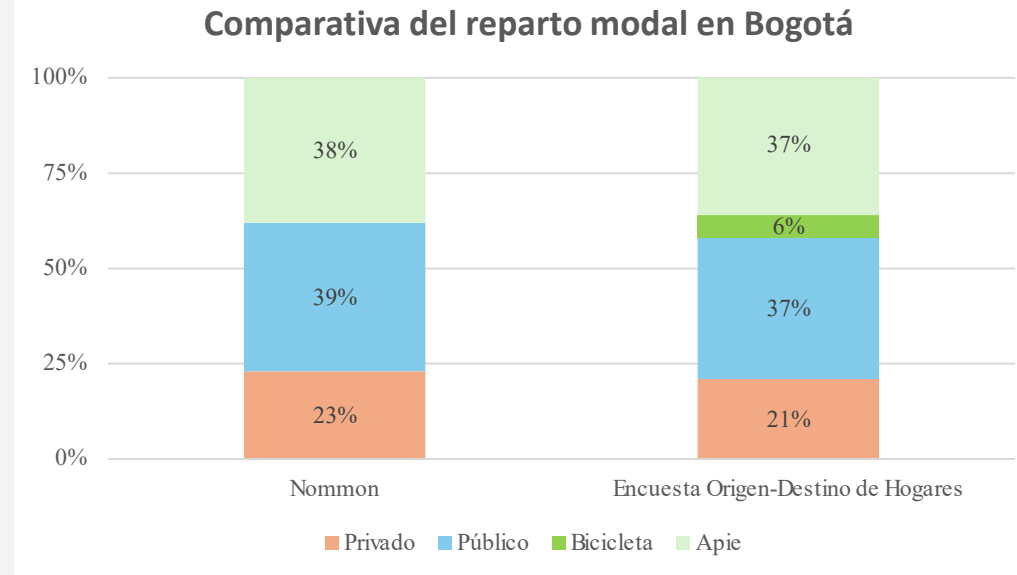
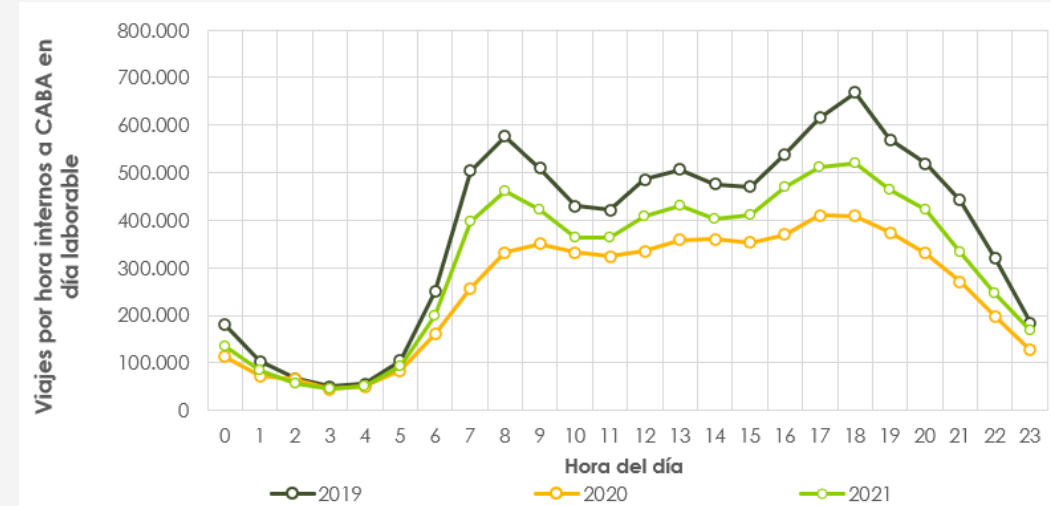
INTERURBAN MOBILITY AT AMAZON REGION (BR+CO)

OBJECTIVE 	Quantification of travel demand in the Brazilian and Colombian Amazon by transport mode: car, river, plane
CLIENT 	World Bank Group
CHALLENGES 	▪ Capture of river transport. ▪ Cross-border trips. ▪ Trips lasting more than 1 day



MOBILITY ANALYSIS AT BUENOS AIRES, BOGOTÁ, MEDELLÍN (AR, CO)

OBJECTIVE 	Estimation of public transport demand in post-COVID Latin America
CLIENT 	Project promoted by the World Bank in collaboration with local governments
CHALLENGES 	- Changes in the general mobility pattern due to the pandemic: changes in work motivations, expansion of e-commerce and teleworking. - Localized changes in public transport demand due to the pandemic. - Data provides insights to guide investments and provide alternatives to the population



KEY TAKEAWAYS

- ✓ Even mature countries lack experience to interpreting data, especially on the public sector
- ✓ Recent movements at Colombia and Mexico indicates a more warm environment to develop projects using MPD as main source of information
- ✓ Chile stands out for its low number of projects, particularly regarding its traditional/classical importance for transport planning
- ✓ Exploiting MPD also means that we need to change the way we see data, new techniques not only on the process of manufacturing the indicator, but also on its application. **More time and training initiatives on LAC region can bust governments** (more averse to innovation) to start accepting this information as a solid resource on bidding process, urban and viability plans



Mobile Phone Data for Policy Program



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Global Data Facility– Mobile Phone Data Program

Goal

Deliver programmatic support to integrate **Mobile Phone Data** into the national data systems of **20 countries by 2030**

Operations Pillar (DEC, Pov, Digital Transformation, ITU)

- **Supports Cohorts of Country Teams** to implement country activities to put MPD to work for Policy Applications and Statistics
- **MPD Foundations Training Program:** workshops and online training programs on MPD for Policy Foundations
- **MPD Essential Resources:** Repository of public goods, including frameworks, tools, use cases, guidelines, content, etc

Research Pillar (DIME, External Board)

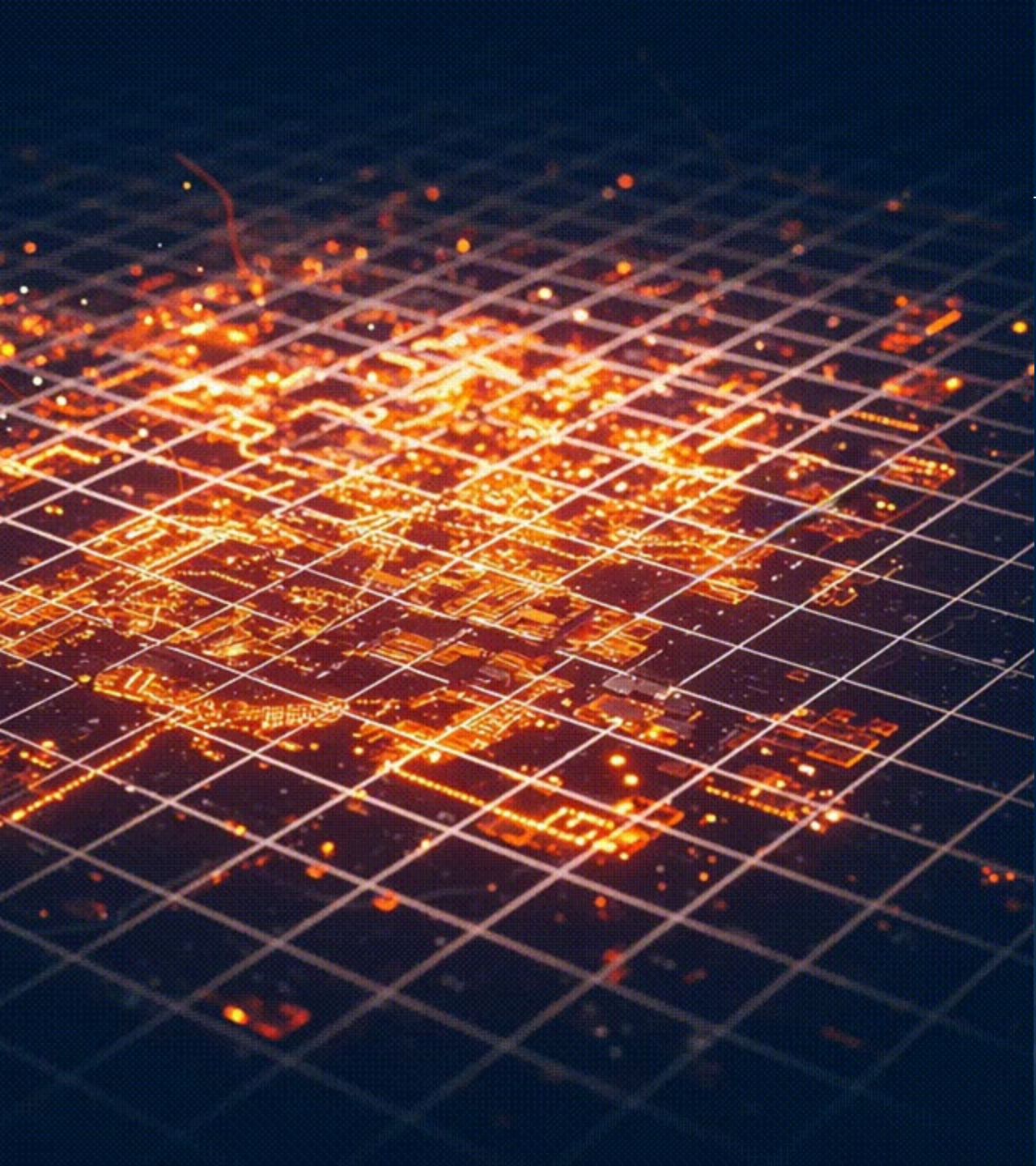
- Innovation in methods and standards at scale
- World class Research Board (Blumenstock, Frias-Martinez, Gonzalez)

Impact



- Supporting Largest Country Cohort to Operationalize MPD for Policy– 30+ Countries
- GDF-MPD grants used to amplify IDA/IBRD operations toward **Integrated National Data Systems**



An aerial night view of a city, likely New York City, with a glowing grid overlay. The grid is composed of white lines that form a perspective grid, and the city lights below are a mix of orange and yellow, creating a vibrant, high-tech aesthetic.

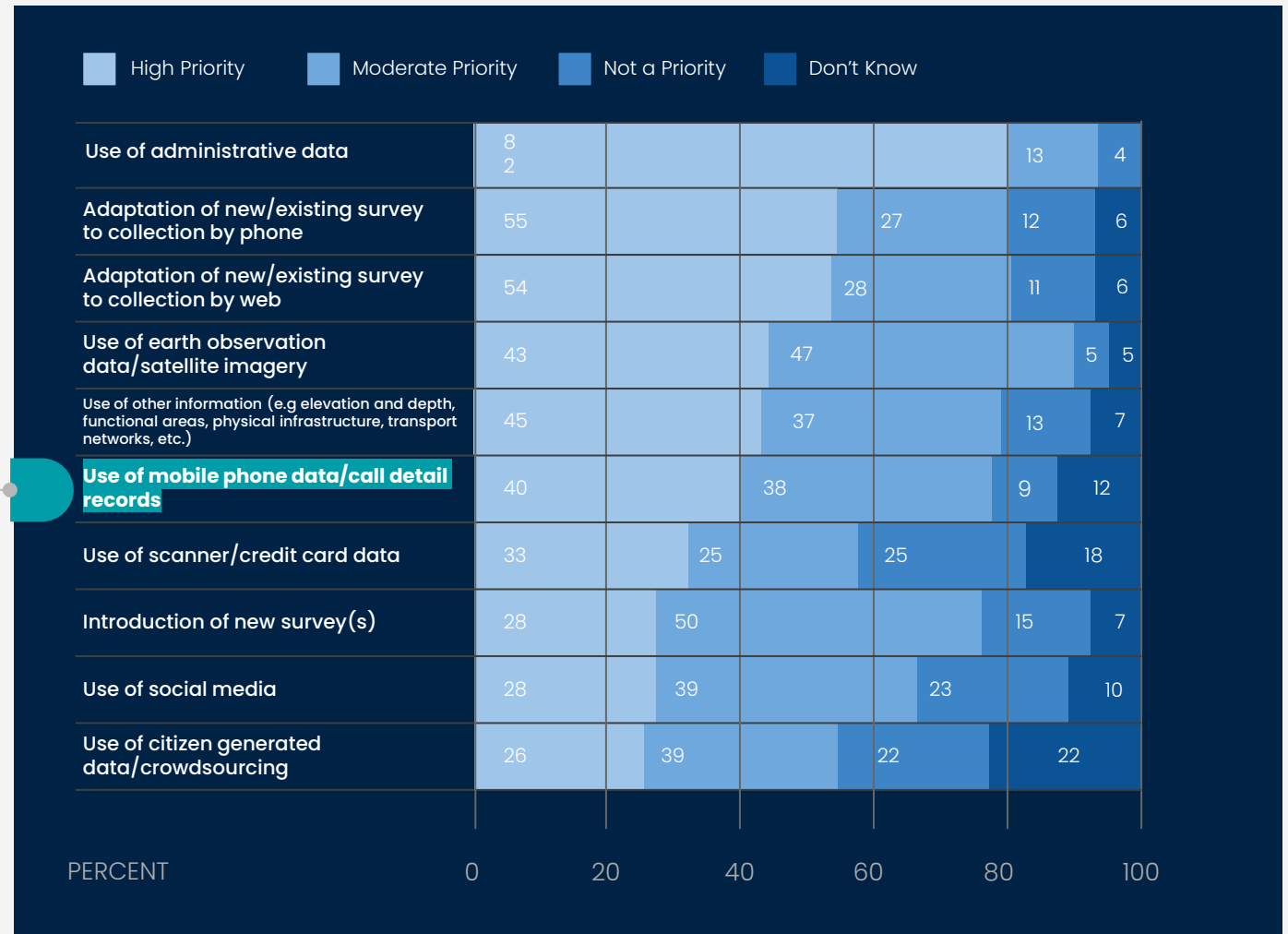
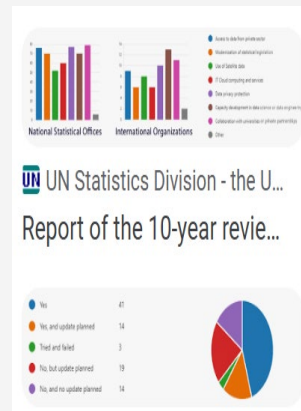
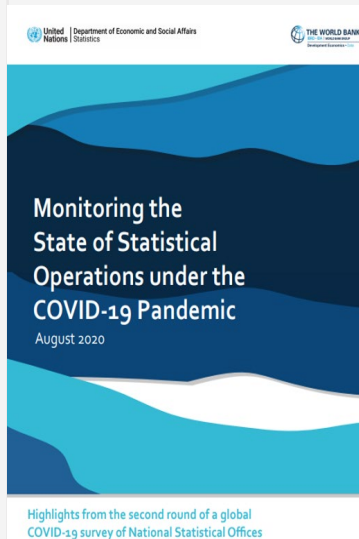
Why invest in **Mobile Phone Big Data** for Policy?

MPD Capacity is a Priority for NSOs

National Statistical Office Priorities for Capacity Development

Question

In the next three years, does your NSO want to expand capacity in the following areas to respond to the new data demands of the 2030 Agenda for sustainable development/SDGs and more recently the COVID-19 pandemic?



78% of National Statistical Offices

consider Mobile Big Data a priority over the next three years (CTAP Round 1)

Less than 20% of NSOs have capacity to use sophisticated data for Policy (World Development Report)

Return **on Impact**

30X

Cost effectiveness. \$30 for every \$1 dollar invested in Integrated National Data Systems (WB ROI report)

Estonia

MPD based Tourism statistics are 2.5x more cost effective; 4x faster than conventional approach

Indonesia

Saved over 1M USD by moving to MPD based tourism statistics

10X

Faster and more cost-effective estimates of small area population dynamics and poverty estimates (Blumenstock)



GDF-MPD Window and Cohort – Based Support Program

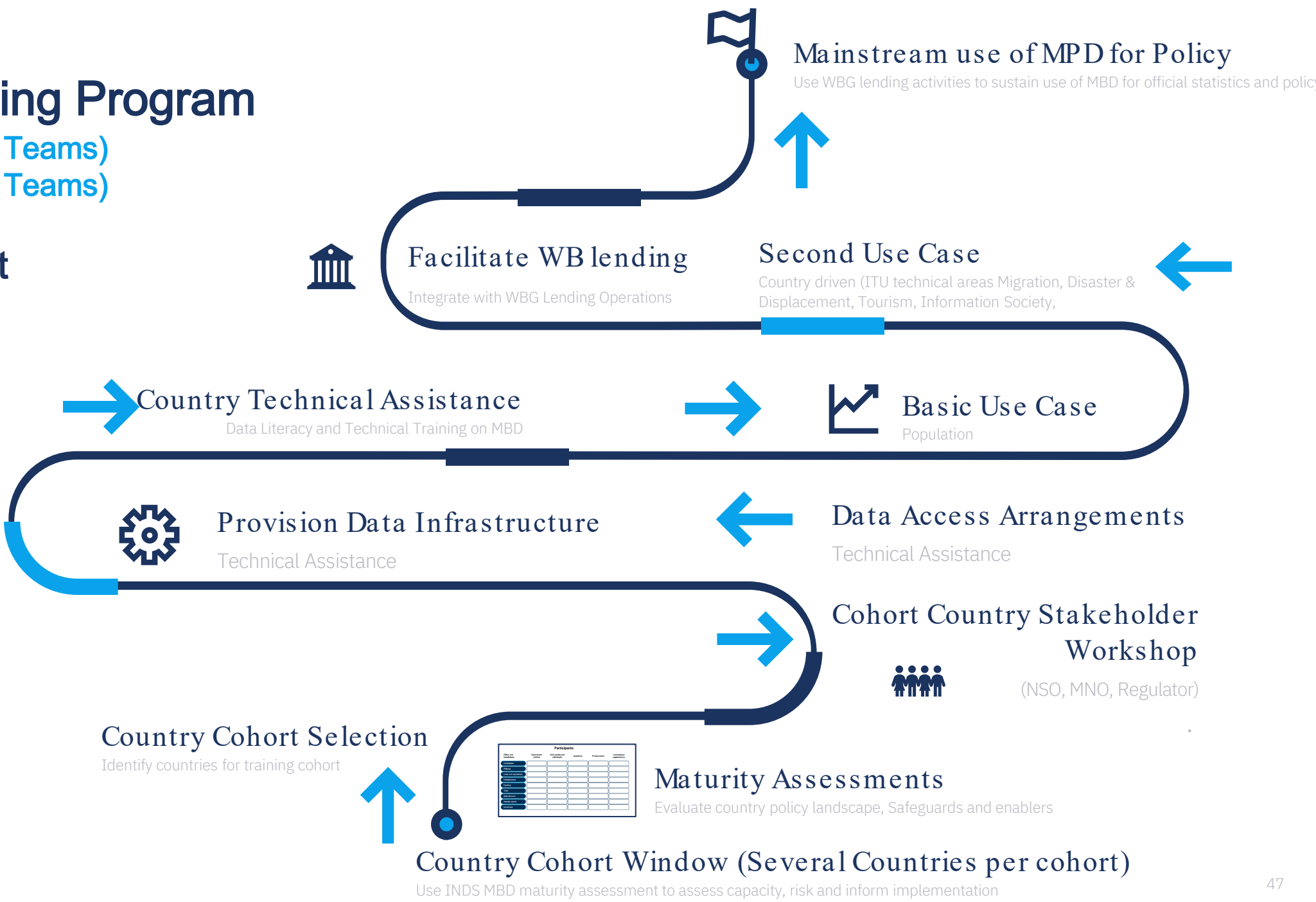
Cohort-Based Capacity Building Program

(Cohort 1: 18 Country Teams)
(Cohort 2: 16 Country Teams)

Practitioner/Expert
Enablers, Standards and Safeguards

Foundation
Enablers, Standards and Safeguards

Pre-Foundation
Awareness Raising

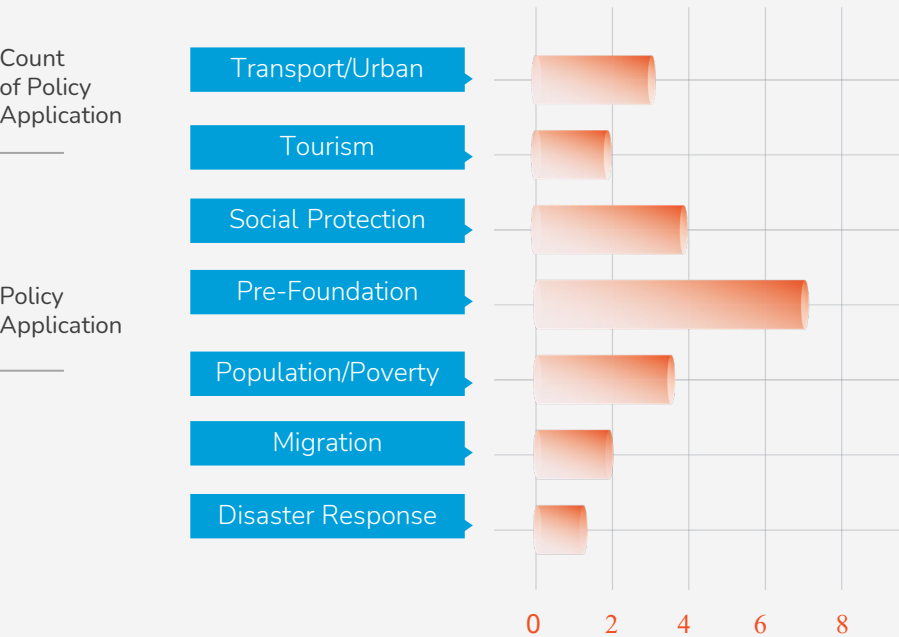


18 Country Teams

2024 Global Cohort



Summary of Cohort Policy Areas



LAC

Argentina | Brazil | Ecuador | Peru | Haiti

AFRICA

Benin | Cote D'Ivoire | Guinea | The Gambia | Senegal | Togo | Tunisia | Morocco | Kenya

ASIA

India | Kazakhstan | Mongolia

16 Country Teams

2025 SADC Cohort



Component 1

- Call for Proposals (Jun)
- Grant Proposal Awarded (Jul)
- 12-18 month implementation

Component 2

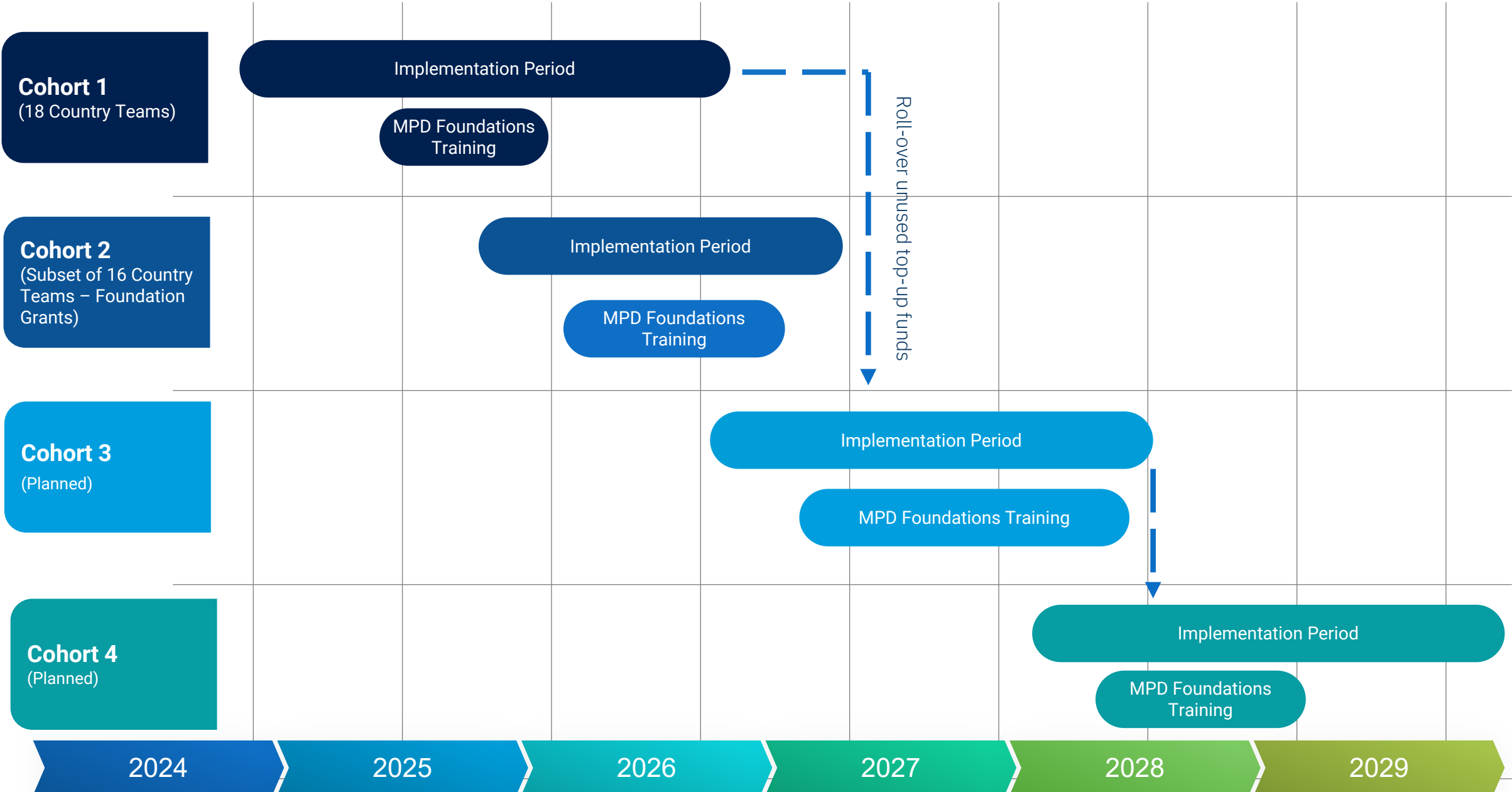
MPD Foundations Training (Oct)

Practitioner Track
Technical Track

Component 3

Successful countries can apply for additional technical assistance via World Bank – SADC Regional Statistics Project

Planned Windows for Country Cohorts



“

If you want to go fast, go alone.

If you want to go far, **go together.**

”





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Thank you



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